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(54) **MINIATURE TELEPHOTO LENS MODULE
AND A CAMERA UTILIZING SUCH A LENS
MODULE**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

2,106,752 A 2/1938 Land
2,354,503 A 7/1944 Arthur
(Continued)

FOREIGN PATENT DOCUMENTS

CN 101025470 A 8/2007
CN 101276415 A 10/2008
(Continued)

OTHER PUBLICATIONS

A compact and cost effective design for cell phone zoom lens,
Chang et al., Sep. 2007, 8 pages.

(Continued)

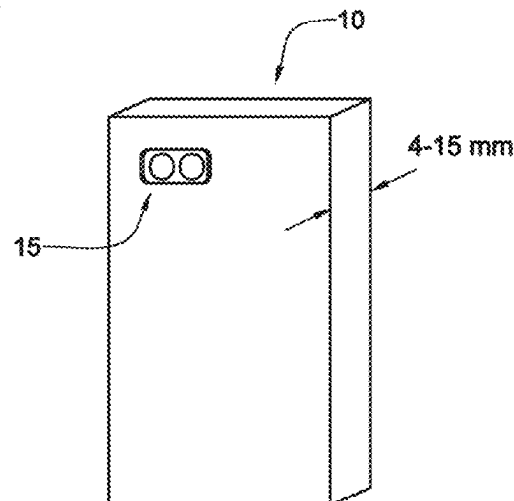
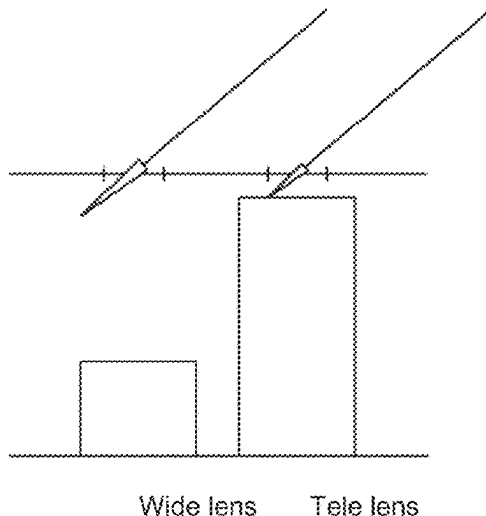
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(57) **ABSTRACT**

The presently disclosed subject matter includes a mobile
electronic comprising an integrated camera, comprising a
Wide camera unit comprising a Wide lens unit, and a
Telephoto camera unit comprising a telephoto lens unit, the
telephoto lens unit and the wide lens unit having respec-
tively TTL/EFL ratios smaller and larger than 1 and defining
separate telephoto and wide optical paths.

19 Claims, 15 Drawing Sheets



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(56)

References Cited

U.S. PATENT DOCUMENTS

2,378,170	A	6/1945	Aklin	6,549,215	B2	4/2003	Jouppi
2,441,093	A	5/1948	Aklin	6,611,289	B1	8/2003	Yu et al.
3,085,354	A	4/1963	Rasmussen et al.	6,643,416	B1	11/2003	Daniels et al.
3,388,956	A	6/1968	Eggert et al.	6,650,368	B1	11/2003	Doron
3,524,700	A	8/1970	Eggert et al.	6,654,180	B2	11/2003	Ori
3,558,218	A	1/1971	Grey	6,680,748	B1	1/2004	Monti
3,584,513	A	6/1971	Gates	6,714,665	B1	3/2004	Hanna et al.
3,864,027	A	2/1975	Harada	6,724,421	B1	4/2004	Glatt
3,941,001	A	3/1976	LaSarge	6,738,073	B2	5/2004	Park et al.
3,942,876	A	3/1976	Betensky	6,741,250	B1	5/2004	Furlan et al.
4,134,645	A	1/1979	Sugiyama et al.	6,750,903	B1	6/2004	Miyatake et al.
4,199,785	A	4/1980	McCullough et al.	6,778,207	B1	8/2004	Lee et al.
4,338,001	A	7/1982	Matsui	7,002,583	B2	2/2006	Rabb, III
4,465,345	A	8/1984	Yazawa	7,015,954	B1	3/2006	Footte et al.
4,792,822	A	12/1988	Akiyama et al.	7,038,716	B2	5/2006	Klein et al.
5,000,551	A	3/1991	Shibayama	7,187,504	B2	3/2007	Horiuchi
5,005,083	A	4/1991	Grage et al.	7,199,348	B2	4/2007	Olsen et al.
5,032,917	A	7/1991	Aschwanden	7,206,136	B2	4/2007	Labaziewicz et al.
5,041,852	A	8/1991	Misawa et al.	7,248,294	B2	7/2007	Slatter
5,051,830	A	9/1991	von Hoessle	7,256,944	B2	8/2007	Labaziewicz et al.
5,099,263	A	3/1992	Matsumoto et al.	7,274,516	B2	9/2007	Kushida et al.
5,248,971	A	9/1993	Mandl	7,305,180	B2	12/2007	Labaziewicz et al.
5,287,093	A	2/1994	Amano et al.	7,339,621	B2	3/2008	Fortier
5,327,291	A	7/1994	Baker et al.	7,346,217	B1	3/2008	Gold, Jr.
5,331,465	A	7/1994	Miyano	7,365,793	B2	4/2008	Cheatle et al.
5,394,520	A	2/1995	Hall	7,411,610	B2	8/2008	Doyle
5,436,660	A	7/1995	Sakamoto	7,424,218	B2	9/2008	Baudisch et al.
5,444,478	A	8/1995	Lelong et al.	7,509,041	B2	3/2009	Hosono
5,459,520	A	10/1995	Sasaki	7,515,351	B2	4/2009	Chen et al.
5,502,537	A	3/1996	Utagawa	7,533,819	B2	5/2009	Barkan et al.
5,600,488	A	2/1997	Minefuji et al.	7,564,635	B1	7/2009	Tang
5,657,402	A	8/1997	Bender et al.	7,619,683	B2	11/2009	Davis
5,682,198	A	10/1997	Katayama et al.	7,643,225	B1	1/2010	Tsai
5,768,443	A	6/1998	Michael et al.	7,660,049	B2	2/2010	Tang
5,892,855	A	4/1999	Kakinami et al.	7,684,128	B2	3/2010	Tang
5,926,190	A	7/1999	Turkowski et al.	7,688,523	B2	3/2010	Sano
5,940,641	A	8/1999	McIntyre et al.	7,692,877	B2	4/2010	Tang et al.
5,969,869	A	10/1999	Hirai et al.	7,697,220	B2	4/2010	Iyama
5,982,951	A	11/1999	Katayama et al.	7,738,016	B2	6/2010	Toyofuku
6,014,266	A	1/2000	Obama et al.	7,738,186	B2	6/2010	Chen et al.
6,035,136	A	3/2000	Hayashi et al.	7,773,121	B1	8/2010	Huntsberger et al.
6,101,334	A	8/2000	Fantone	7,777,972	B1	8/2010	Chen et al.
6,128,416	A	10/2000	Oura	7,809,256	B2	10/2010	Kuroda et al.
6,147,702	A	11/2000	Smith	7,813,057	B2	10/2010	Lin
6,148,120	A	11/2000	Sussman	7,821,724	B2	10/2010	Tang et al.
6,169,636	B1	1/2001	Kreitzer	7,826,149	B2	11/2010	Tang et al.
6,201,533	B1	3/2001	Rosenberg et al.	7,826,151	B2	11/2010	Tsai
6,208,765	B1	3/2001	Bergen	7,869,142	B2	1/2011	Chen et al.
6,211,668	B1	4/2001	Duesler et al.	7,880,776	B2	2/2011	LeGall et al.
6,215,299	B1	4/2001	Reynolds et al.	7,898,747	B2	3/2011	Tang
6,222,359	B1	4/2001	Duesler et al.	7,916,401	B2	3/2011	Chen et al.
6,268,611	B1	7/2001	Pettersson et al.	7,918,398	B2	4/2011	Li et al.
6,320,610	B1	11/2001	Gant et al.	7,957,075	B2	6/2011	Tang
6,341,901	B1	1/2002	Iwasa et al.	7,957,076	B2	6/2011	Tang
6,520,643	B1	2/2003	Holman et al.	7,957,079	B2	6/2011	Tang
				7,961,406	B2	6/2011	Tang et al.
				7,964,835	B2	6/2011	Olsen et al.
				7,978,239	B2	7/2011	Deever et al.
				8,000,031	B1	8/2011	Tsai
				8,004,777	B2	8/2011	Sano et al.
				8,077,400	B2	12/2011	Tang
				8,115,825	B2	2/2012	Culbert et al.
				8,149,327	B2	4/2012	Lin et al.
				8,149,523	B2	4/2012	Ozaki
				8,154,610	B2	4/2012	Jo et al.
				8,218,253	B2	7/2012	Tang
				8,228,622	B2	7/2012	Tang
				8,233,224	B2	7/2012	Chen
				8,238,695	B1	8/2012	Davey et al.
				8,253,843	B2	8/2012	Lin
				8,274,552	B2	9/2012	Dahi et al.
				8,279,537	B2	10/2012	Sato
				8,363,337	B2	1/2013	Tang et al.
				8,390,729	B2	3/2013	Long et al.
				8,391,697	B2	3/2013	Cho et al.
				8,395,851	B2	3/2013	Tang et al.
				8,400,555	B1	3/2013	Georgiev et al.
				8,400,717	B2	3/2013	Chen et al.
				8,439,265	B2	5/2013	Ferren et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

8,446,484 B2	5/2013	Muukki et al.	2003/0030729 A1	2/2003	Prentice et al.
8,451,549 B2	5/2013	Yamanaka et al.	2003/0048542 A1	3/2003	Enomoto
8,483,452 B2	7/2013	Ueda et al.	2003/0093805 A1	5/2003	Gin
8,503,107 B2	8/2013	Chen et al.	2003/0156751 A1	8/2003	Lee et al.
8,514,491 B2	8/2013	Duparre	2003/0160886 A1	8/2003	Misawa et al.
8,514,502 B2	8/2013	Chen	2003/0162564 A1	8/2003	Kimura et al.
8,547,389 B2	10/2013	Hoppe et al.	2003/0202113 A1	10/2003	Yoshikawa
8,553,106 B2	10/2013	Scarff	2004/0008773 A1	1/2004	Itokawa
8,570,668 B2	10/2013	Takakubo et al.	2004/0012683 A1	1/2004	Yamasaki et al.
8,587,691 B2	11/2013	Takane	2004/0017386 A1	1/2004	Liu et al.
8,619,148 B1	12/2013	Watts et al.	2004/0027367 A1	2/2004	Pilu
8,718,458 B2	5/2014	Okuda	2004/0061788 A1	4/2004	Bateman
8,752,969 B1	6/2014	Kane et al.	2004/0095503 A1	5/2004	Iwasawa et al.
8,780,465 B2	7/2014	Chae	2004/0141065 A1	7/2004	Hara et al.
8,803,990 B2	8/2014	Smith	2004/0141086 A1	7/2004	Mihara
8,810,923 B2	8/2014	Shinohara	2004/0169772 A1	9/2004	Matsui et al.
8,854,745 B1	10/2014	Chen	2004/0189849 A1	9/2004	Hofer et al.
8,896,655 B2	11/2014	Mauchly et al.	2004/0227838 A1	11/2004	Atarashi et al.
8,958,164 B2	2/2015	Kwon et al.	2004/0239313 A1	12/2004	Godkin
8,976,255 B2	3/2015	Matsuoto et al.	2004/0240052 A1	12/2004	Minefuji et al.
9,019,387 B2	4/2015	Nakano	2005/0013509 A1	1/2005	Samadani
9,025,073 B2	5/2015	Attar et al.	2005/0041300 A1	2/2005	Oshima et al.
9,025,077 B2	5/2015	Attar et al.	2005/0046740 A1	3/2005	Davis
9,041,835 B2	5/2015	Honda	2005/0062346 A1	3/2005	Sasaki
9,137,447 B2	9/2015	Shibuno	2005/0128604 A1	6/2005	Kuba
9,185,291 B1	11/2015	Shabtay	2005/0134697 A1	6/2005	Mikkonen et al.
9,201,223 B2	12/2015	Ohashi	2005/0141103 A1	6/2005	Nishina
9,215,377 B2	12/2015	Sokeila et al.	2005/0141390 A1	6/2005	Lee et al.
9,215,385 B2	12/2015	Luo	2005/0157184 A1	7/2005	Nakanishi et al.
9,229,194 B2	1/2016	Yoneyama et al.	2005/0168834 A1	8/2005	Matsumoto et al.
9,235,036 B2	1/2016	Kato et al.	2005/0168840 A1	8/2005	Kobayashi et al.
9,270,875 B2	2/2016	Brisedoux et al.	2005/0185049 A1	8/2005	Iwai et al.
9,279,957 B2	3/2016	Kanda et al.	2005/0200718 A1	9/2005	Lee
9,286,680 B1	3/2016	Jiang et al.	2005/0248667 A1	11/2005	Schweng et al.
9,304,305 B1	4/2016	Paul et al.	2005/0270667 A1	12/2005	Gurevich et al.
9,344,626 B2	5/2016	Silverstein et al.	2006/0054782 A1	3/2006	Olsen et al.
9,360,671 B1	6/2016	Zhou	2006/0056056 A1	3/2006	Ahiska et al.
9,369,621 B2	6/2016	Malone et al.	2006/0067672 A1	3/2006	Washisu et al.
9,413,930 B2	8/2016	Geerds	2006/0092524 A1	5/2006	Konno
9,413,984 B2	8/2016	Attar et al.	2006/0102907 A1	5/2006	Lee et al.
9,420,180 B2	8/2016	Jin	2006/0125937 A1	6/2006	LeGall et al.
9,438,792 B2	9/2016	Nakada et al.	2006/0126737 A1	6/2006	Boice et al.
9,485,432 B1	11/2016	Medasani et al.	2006/0170793 A1	8/2006	Pasquarette et al.
9,488,802 B2	11/2016	Chen et al.	2006/0175549 A1	8/2006	Miller et al.
9,568,712 B2	2/2017	Dror et al.	2006/0181619 A1	8/2006	Liow et al.
9,578,257 B2	2/2017	Attar et al.	2006/0187310 A1	8/2006	Janson et al.
9,618,748 B2	4/2017	Munger et al.	2006/0187322 A1	8/2006	Janson et al.
9,678,310 B2	6/2017	Iwasaki et al.	2006/0187338 A1	8/2006	May et al.
9,681,057 B2	6/2017	Attar et al.	2006/0227236 A1	10/2006	Pak
9,723,220 B2	8/2017	Sugie	2006/0238902 A1	10/2006	Nakashima et al.
9,736,365 B2	8/2017	Laroia	2006/0262420 A1	11/2006	Matsumoto et al.
9,736,391 B2	8/2017	Du et al.	2006/0275025 A1	12/2006	Labaziewicz et al.
9,768,310 B2	9/2017	Ahn et al.	2007/0024737 A1	2/2007	Nakamura et al.
9,800,798 B2	10/2017	Ravirala et al.	2007/0035631 A1	2/2007	Ueda
9,817,213 B2	11/2017	Mercado	2007/0077057 A1	4/2007	Chang
9,835,834 B2	12/2017	Li et al.	2007/0114990 A1	5/2007	Godkin
9,851,803 B2	12/2017	Fisher et al.	2007/0126911 A1	6/2007	Nanjo
9,869,846 B1	1/2018	Bone et al.	2007/0127040 A1	6/2007	Davidovici
9,894,287 B2	2/2018	Qian et al.	2007/0159344 A1	7/2007	Kisacanin
9,900,522 B2	2/2018	Lu	2007/0177025 A1	8/2007	Kopet et al.
9,927,600 B2	3/2018	Goldenberg et al.	2007/0183058 A1	8/2007	Bito et al.
10,330,892 B2	6/2019	Hashimoto	2007/0188653 A1	8/2007	Pollock et al.
10,948,696 B2	3/2021	Shabtay et al.	2007/0188884 A1	8/2007	Yoshitsugu et al.
11,340,425 B2	5/2022	Yamazaki	2007/0189386 A1	8/2007	Imagawa et al.
11,347,016 B2	5/2022	Shabtay et al.	2007/0229983 A1	10/2007	Saori
12,069,371 B2	8/2024	Shabtay et al.	2007/0247726 A1	10/2007	Sudoh
2002/0005902 A1	1/2002	Yuen	2007/0253689 A1	11/2007	Nagai et al.
2002/0030163 A1	3/2002	Zhang	2007/0257184 A1	11/2007	Olsen et al.
2002/0054214 A1	5/2002	Yoshikawa	2007/0285550 A1	12/2007	Son
2002/0063711 A1	5/2002	Park et al.	2008/0017557 A1	1/2008	Witdouck
2002/0075258 A1	6/2002	Park et al.	2008/0024614 A1	1/2008	Li et al.
2002/0118471 A1	8/2002	Imoto	2008/0025634 A1	1/2008	Border et al.
2002/0122113 A1	9/2002	Foote	2008/0030592 A1	2/2008	Border et al.
2002/0136554 A1	9/2002	Nomura et al.	2008/0030611 A1	2/2008	Jenkins
2002/0167741 A1	11/2002	Koiwai et al.	2008/0056698 A1	3/2008	Lee et al.
			2008/0084484 A1	4/2008	Ochi et al.
			2008/0088942 A1	4/2008	Seo
			2008/0094730 A1	4/2008	Toma et al.
			2008/0094738 A1	4/2008	Lee

(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0106629	A1	5/2008	Kurtz et al.	2011/0221599	A1	9/2011	Högasten
2008/0117316	A1	5/2008	Orimoto	2011/0229054	A1	9/2011	Weston et al.
2008/0117527	A1	5/2008	Nuno et al.	2011/0234798	A1	9/2011	Chou
2008/0129831	A1	6/2008	Cho et al.	2011/0234853	A1	9/2011	Hayashi et al.
2008/0218611	A1	9/2008	Parulski et al.	2011/0234881	A1	9/2011	Wakabayashi et al.
2008/0218612	A1	9/2008	Border et al.	2011/0242286	A1	10/2011	Pace et al.
2008/0218613	A1	9/2008	Janson et al.	2011/0242355	A1	10/2011	Goma et al.
2008/0219654	A1	9/2008	Border et al.	2011/0249347	A1	10/2011	Kubota
2008/0273250	A1	11/2008	Nishio	2011/0285714	A1	11/2011	Swic et al.
2008/0291531	A1	11/2008	Heimer	2011/0298966	A1	12/2011	Kirschstein et al.
2008/0304161	A1	12/2008	Souma	2011/0310219	A1	12/2011	Kim et al.
2009/0002839	A1	1/2009	Sato	2012/0014682	A1	1/2012	David et al.
2009/0067063	A1	3/2009	Asami et al.	2012/0026366	A1	2/2012	Golan et al.
2009/0086074	A1	4/2009	Li et al.	2012/0044372	A1	2/2012	Cote et al.
2009/0102948	A1	4/2009	Scherling	2012/0062780	A1	3/2012	Morihisa
2009/0109556	A1	4/2009	Shimizu et al.	2012/0062783	A1	3/2012	Tang et al.
2009/0122195	A1	5/2009	Van Baar et al.	2012/0069235	A1	3/2012	Imai
2009/0122406	A1	5/2009	Rouvinen et al.	2012/0069455	A1	3/2012	Lin et al.
2009/0122423	A1	5/2009	Park et al.	2012/0075489	A1	3/2012	Nishihara
2009/0128644	A1	5/2009	Camp et al.	2012/0092777	A1	4/2012	Tochigi et al.
2009/0135245	A1	5/2009	Luo et al.	2012/0098927	A1	4/2012	Sablak et al.
2009/0141365	A1	6/2009	Jannard et al.	2012/0105579	A1	5/2012	Jeon et al.
2009/0147368	A1	6/2009	Oh et al.	2012/0105708	A1	5/2012	Hagiwara
2009/0161228	A1	6/2009	Lee	2012/0124525	A1	5/2012	Kang
2009/0168135	A1	7/2009	Yu et al.	2012/0147489	A1	6/2012	Matsuoka
2009/0190909	A1	7/2009	Mise et al.	2012/0154547	A1	6/2012	Aizawa
2009/0200451	A1	8/2009	Connors	2012/0154614	A1	6/2012	Moriya et al.
2009/0219547	A1	9/2009	Kauhanen et al.	2012/0154929	A1	6/2012	Tsai et al.
2009/0225438	A1	9/2009	Kubota	2012/0194923	A1	8/2012	Um
2009/0234542	A1	9/2009	Orlewski	2012/0196648	A1	8/2012	Havens et al.
2009/0252484	A1	10/2009	Hasuda et al.	2012/0229663	A1	9/2012	Nelson et al.
2009/0279191	A1	11/2009	Yu	2012/0229920	A1	9/2012	Otsu et al.
2009/0295949	A1	12/2009	Ojala	2012/0249815	A1	10/2012	Bohn et al.
2009/0295986	A1	12/2009	Topliss et al.	2012/0262806	A1	10/2012	Lin et al.
2009/0303620	A1	12/2009	Abe et al.	2012/0287315	A1	11/2012	Huang et al.
2009/0324135	A1	12/2009	Kondo et al.	2012/0314299	A1	12/2012	Tashiro et al.
2010/0007967	A1	1/2010	Ohashi	2012/0320467	A1	12/2012	Baik et al.
2010/0013906	A1	1/2010	Border et al.	2013/0002928	A1	1/2013	Imai
2010/0020221	A1	1/2010	Tupman et al.	2013/0002933	A1	1/2013	Topliss et al.
2010/0026878	A1	2/2010	Seo	2013/0016427	A1	1/2013	Sugawara
2010/0033844	A1	2/2010	Katano	2013/0057971	A1	3/2013	Zhao et al.
2010/0060746	A9	3/2010	Olsen et al.	2013/0063629	A1	3/2013	Webster et al.
2010/0060995	A1	3/2010	Yumiki et al.	2013/0076922	A1	3/2013	Shihoh et al.
2010/0097444	A1	4/2010	Lablans	2013/0088788	A1	4/2013	You
2010/0103194	A1	4/2010	Chen et al.	2013/0093842	A1	4/2013	Yahata
2010/0134621	A1	6/2010	Namkoong et al.	2013/0094126	A1	4/2013	Rappoport et al.
2010/0165131	A1	7/2010	Makimoto et al.	2013/0113894	A1	5/2013	Mirlay
2010/0165476	A1	7/2010	Eguchi	2013/0135445	A1	5/2013	Dahi et al.
2010/0196001	A1	8/2010	Ryynänen et al.	2013/0148215	A1	6/2013	Mori et al.
2010/0202068	A1	8/2010	Ito	2013/0148854	A1	6/2013	Wang et al.
2010/0214664	A1	8/2010	Chia	2013/0155176	A1	6/2013	Paripally et al.
2010/0238327	A1	9/2010	Griffith et al.	2013/0163085	A1	6/2013	Lim et al.
2010/0246024	A1	9/2010	Aoki et al.	2013/0176479	A1	7/2013	Wada
2010/0259836	A1	10/2010	Kang et al.	2013/0182150	A1	7/2013	Asakura
2010/0265331	A1	10/2010	Tanaka	2013/0201360	A1	8/2013	Song
2010/0277813	A1	11/2010	Ito	2013/0202273	A1	8/2013	Ouedraogo et al.
2010/0283842	A1	11/2010	Guissin et al.	2013/0208178	A1	8/2013	Park
2010/0321494	A1	12/2010	Peterson et al.	2013/0229544	A1	9/2013	Bando
2011/0001838	A1	1/2011	Lee	2013/0235224	A1	9/2013	Park et al.
2011/0032409	A1	2/2011	Rossi et al.	2013/0250150	A1	9/2013	Malone et al.
2011/0058320	A1	3/2011	Kim et al.	2013/0258044	A1	10/2013	Betts-Lacroix
2011/0063417	A1	3/2011	Peters et al.	2013/0258048	A1	10/2013	Wang et al.
2011/0063446	A1	3/2011	McMordie et al.	2013/0270419	A1	10/2013	Singh et al.
2011/0064327	A1	3/2011	Dagher et al.	2013/0271852	A1	10/2013	Schuster
2011/0080487	A1	4/2011	Venkataraman et al.	2013/0278785	A1	10/2013	Nomura et al.
2011/0080655	A1	4/2011	Mori	2013/0279032	A1	10/2013	Sugetsu et al.
2011/0102667	A1	5/2011	Chua et al.	2013/0286221	A1	10/2013	Shechtman et al.
2011/0102911	A1	5/2011	Iwasaki	2013/0286488	A1	10/2013	Chae
2011/0115965	A1	5/2011	Engelhardt et al.	2013/0321668	A1	12/2013	Kamath
2011/0121666	A1	5/2011	Park et al.	2014/0009631	A1	1/2014	Topliss
2011/0128288	A1	6/2011	Petrou et al.	2014/0022436	A1	1/2014	Kim et al.
2011/0149119	A1	6/2011	Matsui	2014/0036112	A1	2/2014	Scarff
2011/0157430	A1	6/2011	Hosoya et al.	2014/0049615	A1	2/2014	Uwagawa
2011/0164172	A1	7/2011	Shintani et al.	2014/0063616	A1	3/2014	Okano et al.
2011/0188121	A1	8/2011	Goring et al.	2014/0092487	A1	4/2014	Chen et al.
				2014/0118584	A1	5/2014	Lee et al.
				2014/0139719	A1	5/2014	Fukaya et al.
				2014/0146216	A1	5/2014	Okumura
				2014/0160311	A1	6/2014	Hwang et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2014/0160581 A1	6/2014	Cho et al.		2016/0212333 A1	7/2016	Liege et al.
2014/0192224 A1	7/2014	Laroia		2016/0212358 A1	7/2016	Shikata
2014/0192238 A1	7/2014	Attar et al.		2016/0212418 A1	7/2016	Demirdjian et al.
2014/0192253 A1	7/2014	Laroia		2016/0238834 A1	8/2016	Erlich et al.
2014/0204480 A1 *	7/2014	Jo	G02B 9/60 359/764	2016/0241751 A1	8/2016	Park
2014/0218587 A1	8/2014	Shah		2016/0241756 A1	8/2016	Chen
2014/0240853 A1	8/2014	Kubota et al.		2016/0245669 A1	8/2016	Nomura
2014/0285907 A1	9/2014	Tang et al.		2016/0291295 A1	10/2016	Shabtay
2014/0293453 A1	10/2014	Ogino et al.		2016/0295112 A1	10/2016	Georgiev et al.
2014/0313316 A1	10/2014	Olsson et al.		2016/0301840 A1	10/2016	Du et al.
2014/0362242 A1	12/2014	Takizawa		2016/0301868 A1	10/2016	Acharya et al.
2014/0362274 A1	12/2014	Christie et al.		2016/0306161 A1	10/2016	Harada et al.
2014/0376090 A1	12/2014	Terajima		2016/0313537 A1	10/2016	Mercado
2014/0379103 A1	12/2014	Ishikawa et al.		2016/0341931 A1	11/2016	Liu et al.
2015/0002683 A1	1/2015	Hu et al.		2016/0342095 A1	11/2016	Bieling et al.
2015/0002684 A1	1/2015	Kuchiki		2016/0349504 A1	12/2016	Kim et al.
2015/0022896 A1	1/2015	Cho et al.		2016/0353008 A1	12/2016	Osborne
2015/0029601 A1	1/2015	Dror et al.		2016/0353012 A1	12/2016	Kao et al.
2015/0042870 A1	2/2015	Chan et al.		2016/0381289 A1	12/2016	Kim et al.
2015/0070781 A1	3/2015	Cheng et al.		2017/0001577 A1	1/2017	Seagraves et al.
2015/0086127 A1	3/2015	Camilus et al.		2017/0019616 A1	1/2017	Zhu et al.
2015/0092066 A1	4/2015	Geiss et al.		2017/0023778 A1	1/2017	Inoue
2015/0103147 A1	4/2015	Ho et al.		2017/0052350 A1	2/2017	Chen
2015/0110345 A1	4/2015	Weichselbaum		2017/0070731 A1	3/2017	Darling et al.
2015/0116569 A1	4/2015	Mercado		2017/0094187 A1	3/2017	Sharma et al.
2015/0124059 A1	5/2015	Georgiev et al.		2017/0102522 A1	4/2017	Jo
2015/0138381 A1	5/2015	Ahn		2017/0115466 A1	4/2017	Murakami et al.
2015/0138431 A1	5/2015	Shin et al.		2017/0115471 A1	4/2017	Shinohara
2015/0145965 A1	5/2015	Livvatan et al.		2017/0124987 A1	5/2017	Kim et al.
2015/0153548 A1	6/2015	Kim et al.		2017/0150061 A1	5/2017	Shabtay et al.
2015/0154776 A1	6/2015	Zhang et al.		2017/0153422 A1	6/2017	Tang et al.
2015/0160438 A1	6/2015	Okuda		2017/0160511 A1	6/2017	Kim et al.
2015/0162048 A1	6/2015	Hirata et al.		2017/0176711 A1	6/2017	Iwasaki et al.
2015/0168667 A1	6/2015	Kudoh		2017/0187962 A1	6/2017	Lee et al.
2015/0177496 A1	6/2015	Marks et al.		2017/0199360 A1	7/2017	Chang
2015/0181115 A1	6/2015	Mashiah		2017/0214846 A1	7/2017	Du et al.
2015/0195458 A1	7/2015	Nakayama et al.		2017/0214866 A1	7/2017	Zhu et al.
2015/0198464 A1	7/2015	El Alami		2017/0219749 A1	8/2017	Hou et al.
2015/0205068 A1	7/2015	Sasaki		2017/0230552 A1	8/2017	Eromaki et al.
2015/0215516 A1	7/2015	Dolgin		2017/0242225 A1	8/2017	Fiske
2015/0237280 A1	8/2015	Choi et al.		2017/0276911 A1	9/2017	Huang
2015/0242994 A1	8/2015	Shen		2017/0276914 A1	9/2017	Yao et al.
2015/0244906 A1	8/2015	Wu et al.		2017/0276954 A1	9/2017	Bajorins et al.
2015/0244942 A1 *	8/2015	Shabtay	G02B 9/60 29/832	2017/0289458 A1	10/2017	Song et al.
2015/0253532 A1	9/2015	Lin		2017/0294002 A1	10/2017	Jia et al.
2015/0253543 A1	9/2015	Mercado		2017/0310952 A1	10/2017	Adomat et al.
2015/0253647 A1	9/2015	Mercado		2017/0329108 A1	11/2017	Hashimoto
2015/0261299 A1	9/2015	Wajs		2017/0329111 A1	11/2017	Hu et al.
2015/0271471 A1	9/2015	Hsieh et al.		2017/0337703 A1	11/2017	Wu et al.
2015/0281678 A1	10/2015	Park et al.		2018/0003925 A1	1/2018	Shmunk
2015/0286033 A1	10/2015	Osborne		2018/0013944 A1	1/2018	Evans, V et al.
2015/0288865 A1	10/2015	Osborne		2018/0017844 A1	1/2018	Yu et al.
2015/0296112 A1	10/2015	Park et al.		2018/0024319 A1	1/2018	Lai et al.
2015/0316744 A1	11/2015	Chen		2018/0024329 A1	1/2018	Goldenberg et al.
2015/0323757 A1	11/2015	Bone		2018/0048825 A1	2/2018	Wang
2015/0334309 A1	11/2015	Peng et al.		2018/0059365 A1	3/2018	Bone et al.
2015/0373252 A1	12/2015	Georgiev		2018/0059376 A1	3/2018	Lin et al.
2015/0373263 A1	12/2015	Georgiev et al.		2018/0059379 A1	3/2018	Chou
2016/0007008 A1	1/2016	Molgaard et al.		2018/0081149 A1	3/2018	Bae et al.
2016/0028949 A1	1/2016	Lee et al.		2018/0109660 A1	4/2018	Yoon et al.
2016/0033742 A1	2/2016	Huang		2018/0109710 A1	4/2018	Lee et al.
2016/0044250 A1	2/2016	Shabtay et al.		2018/0120674 A1	5/2018	Avivi et al.
2016/0062084 A1	3/2016	Chen et al.		2018/0149835 A1	5/2018	Park
2016/0062136 A1	3/2016	Nomura et al.		2018/0150973 A1	5/2018	Tang et al.
2016/0070088 A1	3/2016	Koguchi		2018/0176426 A1	6/2018	Wei et al.
2016/0085089 A1	3/2016	Mercado		2018/0183982 A1	6/2018	Lee et al.
2016/0105616 A1	4/2016	Shabtay et al.		2018/0184010 A1	6/2018	Cohen et al.
2016/0154066 A1	6/2016	Hioka et al.		2018/0196236 A1	7/2018	Ohashi et al.
2016/0154202 A1	6/2016	Wippermann et al.		2018/0196238 A1	7/2018	Goldenberg et al.
2016/0154204 A1	6/2016	Lim et al.		2018/0198897 A1	7/2018	Tang et al.
2016/0187631 A1	6/2016	Choi et al.		2018/0216925 A1	8/2018	Yasuda et al.
2016/0195691 A1	7/2016	Bitto et al.		2018/0217475 A1	8/2018	Goldenberg et al.
2016/0202455 A1	7/2016	Aschwanden et al.		2018/0218224 A1	8/2018	Olmstead et al.
				2018/0224630 A1	8/2018	Lee et al.
				2018/0241922 A1	8/2018	Baldwin et al.
				2018/0249090 A1	8/2018	Nakagawa et al.
				2018/0253877 A1	9/2018	Kozub et al.
				2018/0268226 A1	9/2018	Shashua et al.
				2018/0295292 A1	10/2018	Lee et al.

(56)

References Cited

FOREIGN PATENT DOCUMENTS

U.S. PATENT DOCUMENTS					
			CN	101634738 A	1/2010
			CN	201514511 U	6/2010
2018/0300901 A1	10/2018	Wakai et al.	CN	102130567 A	7/2011
2018/0307005 A1	10/2018	Price et al.	CN	102147519 A	8/2011
2018/0329281 A1	11/2018	Ye	CN	102193162 A	9/2011
2018/0368656 A1	12/2018	Austin et al.	CN	102215373 A	10/2011
2019/0025549 A1	1/2019	Hsueh et al.	CN	102466865 A	5/2012
2019/0025554 A1	1/2019	Son	CN	102466867 A	5/2012
2019/0049687 A1	2/2019	Bachar et al.	CN	102739949 A	10/2012
2019/0075284 A1	3/2019	Ono	CN	102147519 B	1/2013
2019/0086638 A1	3/2019	Lee	CN	102982518 A	3/2013
2019/0089941 A1	3/2019	Bigioi et al.	CN	103024272 A	4/2013
2019/0094500 A1	3/2019	Tseng et al.	CN	203406908 U	1/2014
2019/0096047 A1	3/2019	Ogasawara	CN	103576290 A	2/2014
2019/0100156 A1	4/2019	Chung et al.	CN	203482298 U	3/2014
2019/0107651 A1	4/2019	Sade	CN	103698876 A	4/2014
2019/0121103 A1	4/2019	Bachar et al.	CN	103841404 A	6/2014
2019/0121216 A1	4/2019	Shabtay et al.	CN	104297906 A	1/2015
2019/0130822 A1	5/2019	Jung et al.	CN	104407432 A	3/2015
2019/0154466 A1	5/2019	Fletcher	CN	204422947 U	6/2015
2019/0155002 A1	5/2019	Shabtay et al.	CN	105467563 A	4/2016
2019/0170965 A1	6/2019	Shabtay	CN	105657290 A	6/2016
2019/0187443 A1	6/2019	Jia et al.	CN	205301703 U	6/2016
2019/0187486 A1	6/2019	Goldenberg et al.	CN	105827903 A	8/2016
2019/0196148 A1	6/2019	Yao et al.	CN	105847662 A	8/2016
2019/0213712 A1	7/2019	Lashdan et al.	CN	105872325 A	8/2016
2019/0215440 A1	7/2019	Rivard et al.	CN	106680974 A	5/2017
2019/0222758 A1	7/2019	Goldenberg et al.	CN	104570280 B	6/2017
2019/0227338 A1	7/2019	Bachar et al.	CN	107608052 A	1/2018
2019/0228562 A1	7/2019	Song	CN	107682489 A	2/2018
2019/0235202 A1	8/2019	Smyth et al.	CN	105467563 B	2/2019
2019/0297238 A1	9/2019	Klosterman	CN	109729266 A	5/2019
2019/0320119 A1	10/2019	Miyoshi	CN	111988454 A	11/2020
2019/0353874 A1	11/2019	Yeh et al.	CN	112764200 A	5/2021
2020/0014912 A1	1/2020	Kytsun et al.	EP	1536633 A1	6/2005
2020/0064597 A1	2/2020	Shabtay et al.	EP	1780567 A1	5/2007
2020/0084358 A1	3/2020	Nadamoto	EP	2523450 A1	11/2012
2020/0092486 A1	3/2020	Guo et al.	JP	S54157620 A	12/1979
2020/0103726 A1	4/2020	Shabtay et al.	JP	S59121015 A	7/1984
2020/0104034 A1	4/2020	Lee et al.	JP	S59191146 A	10/1984
2020/0118287 A1	4/2020	Hsieh et al.	JP	6165212 A	4/1986
2020/0134848 A1	4/2020	El-Khamy et al.	JP	S6370211 A	3/1988
2020/0162682 A1	5/2020	Cheng et al.	JP	H0233117 A	2/1990
2020/0192069 A1	6/2020	Makeev et al.	JP	04211230 A	8/1992
2020/0220956 A1	7/2020	Fujisaki et al.	JP	406059195 A	3/1994
2020/0221026 A1	7/2020	Fridman et al.	JP	H06258702 A	9/1994
2020/0241233 A1	7/2020	Shabtay et al.	JP	H06347687 A	12/1994
2020/0264403 A1	8/2020	Bachar et al.	JP	H07120673 A	5/1995
2020/0314224 A1	10/2020	Yang	JP	H07318864 A	12/1995
2020/0333691 A1	10/2020	Shabtay et al.	JP	H07325246 A	12/1995
2020/0389580 A1	12/2020	Kodama et al.	JP	H07333505 A	12/1995
2020/0400926 A1	12/2020	Bachar	JP	H08179215 A	7/1996
2021/0026117 A1	1/2021	Yao	JP	08271976 A	10/1996
2021/0048628 A1	2/2021	Shabtay et al.	JP	H09211326 A	8/1997
2021/0048649 A1	2/2021	Goldenberg et al.	JP	H114373 A	1/1999
2021/0165192 A1	6/2021	Goldenberg et al.	JP	H11223771 A	8/1999
2021/0180989 A1	6/2021	Fukumura et al.	JP	2000131610 A	5/2000
2021/0208415 A1	7/2021	Goldenberg et al.	JP	2000292848 A	10/2000
2021/0263276 A1	8/2021	Huang et al.	JP	3210242 B2	9/2001
2021/0333521 A9	10/2021	Yedid et al.	JP	2002010276 A	1/2002
2021/0364746 A1	11/2021	Chen	JP	2002365549 A	12/2002
2021/0368104 A1	11/2021	Bian et al.	JP	2003298920 A	10/2003
2021/0396974 A1	12/2021	Kuo	JP	2003304024 A	10/2003
2022/0004085 A1	1/2022	Shabtay et al.	JP	2003329932 A	11/2003
2022/0046151 A1	2/2022	Shabtay et al.	JP	2004056779 A	2/2004
2022/0066168 A1	3/2022	Shi	JP	2004133054 A	4/2004
2022/0113511 A1	4/2022	Chen	JP	2004226563 A	8/2004
2022/0146910 A1	5/2022	Li et al.	JP	2004245982 A	9/2004
2022/0206264 A1	6/2022	Rudnick et al.	JP	2004247887 A	9/2004
2022/0232167 A1	7/2022	Shabtay et al.	JP	2004334185 A	11/2004
2022/0252963 A1	8/2022	Shabtay et al.	JP	2005099265 A	4/2005
2022/0368814 A1	11/2022	Topliss et al.	JP	2005122084 A	5/2005
2023/0022701 A1	1/2023	Li et al.	JP	2005321592 A	11/2005
2023/0080199 A1	3/2023	Eromaki et al.	JP	2006038891 A	2/2006
			JP	2006191411 A	7/2006
			JP	2006195139 A	7/2006
			JP	2006237914 A	9/2006

(56)

References Cited

FOREIGN PATENT DOCUMENTS

JP	2006238325	A	9/2006
JP	2008083377	A	9/2006
JP	2007086808	A	4/2007
JP	2007133096	A	5/2007
JP	2007164065	A	6/2007
JP	2007219199	A	8/2007
JP	2007228006	A	9/2007
JP	2007306282	A	11/2007
JP	2008076485	A	4/2008
JP	2008111876	A	5/2008
JP	2008191423	A	8/2008
JP	2008245142	A	10/2008
JP	2008271026	A	11/2008
JP	2010032936	A	2/2010
JP	2010164841	A	7/2010
JP	2010204341	A	9/2010
JP	2011055246	A	3/2011
JP	2011085666	A	4/2011
JP	2011138407	A	7/2011
JP	2011145315	A	7/2011
JP	2011151448	A	8/2011
JP	2011203283	A	10/2011
JP	2012132739	A	7/2012
JP	2012203234	A	10/2012
JP	2012230323	A	11/2012
JP	2013003317	A	1/2013
JP	2013003754	A	1/2013
JP	2013101213	A	5/2013
JP	2013105049	A	5/2013
JP	2013106289	A	5/2013
JP	2013148823	A	8/2013
JP	2014142542	A	8/2014
JP	2016105577	A	6/2016
JP	2017116679	A	6/2017
JP	2017146440	A	8/2017
JP	2018022123	A	2/2018
JP	2018059969	A	4/2018
JP	2019028249	A	2/2019
JP	2019113878	A	7/2019
JP	2019126179	A	7/2019
KR	20070005946	A	1/2007
KR	20080088477	A	10/2008
KR	20090019525	A	2/2009
KR	20090058229	A	6/2009
KR	20090131805	A	12/2009
KR	20100008936	A	1/2010
KR	20110058094	A	6/2011
KR	20110080590	A	7/2011
KR	20110082494	A	7/2011
KR	20110115391	A	10/2011
KR	20120068177	A	6/2012
KR	20140135909	A	5/2013
KR	20130104764	A	9/2013
KR	1020130135805	A	11/2013
KR	20140014787	A	2/2014
KR	20140023552	A	2/2014
KR	101428042	B1	8/2014
KR	101477178	B1	12/2014
KR	20140144126	A	12/2014
KR	20150118012	A	10/2015
KR	20160000759	A	1/2016
KR	101632168	B1	6/2016
KR	20160115359	A	10/2016
KR	20170105236	A	9/2017
KR	20180120894	A	11/2018
KR	20130085116	A	6/2019
KR	1020200005332	A	1/2020
TW	1407177	B	9/2013
TW	M602642	U	10/2020
WO	2000027131	A2	5/2000
WO	2004084542	A1	9/2004
WO	2006008805	A1	1/2006
WO	2010122841	A1	10/2010

WO	2013058111	A1	4/2013
WO	2013063097	A1	5/2013
WO	2014072818	A2	5/2014
WO	2017025822	A1	2/2017
WO	2017037688	A1	3/2017
WO	2018130898	A1	7/2018

OTHER PUBLICATIONS

Consumer Electronic Optics: How small a lens can be? The case of panomorph lenses, Thibault et al., Sep. 2014, 7 pages.

Optical design of camera optics for mobile phones, Steinich et al., 2012, pp. 51-58 (8 pages).

The Optics of Miniature Digital Camera Modules, Bareau et al., 2006, 11 pages.

Modeling and measuring liquid crystal tunable lenses, Peter P. Clark, 2014, 7 pages.

Mobile Platform Optical Design, Peter P. Clark, 2014, 7 pages.

Boye et al., "Ultrathin Optics for Low-Profile Innocuous Imager", Sandia Report, 2009, pp. 56-56.

"Cheat sheet: how to understand f-stops", Internet article, Digital Camera World, 2017.

Zitova B et al: "Image Registration Methods: a Survey", Image and Vision Computing, Elsevier, Guildford, GB, vol. 21, No. 11, Oct. 1, 2003 (Oct. 1, 2003), pp. 977-1000, XP00i 189327, ISSN: 0262-8856, DOI: 10.1016/S0262-8856(03)00137-9.

Itay Yedid: "The Evolution of Zoom Camera Technologies in Smartphones", Corephotonics White Paper, Aug. 1, 2017 (Aug. 1, 2017), XP055980796.

George B Arfken: "Mathematical Methods for Physicists: A Comprehensive Guide" In: "Mathematical Methods for Physicists: A Comprehensive Guide", Jan. 1, 2013 (Jan. 1, 2013), Elsevier, XP093159030, ISBN: 978-0-12-384654-9 pp. 195-196.

Statistical Modeling and Performance Characterization of a Real-Time Dual Camera Surveillance System, Greienhagen et al., Publisher: IEEE, 2000, 8 pages.

A 3MPixel Multi-Aperture Image Sensor with 0.7µm Pixels in 0.11µm CMOS, Fife et al., Stanford University, 2008, 3 pages.

Dual camera intelligent sensor for high definition 360 degrees surveillance, Scotti et al., Publisher: IET, May 9, 2000, 8 pages.

Dual-sensor foveated imaging system, Hua et al., Publisher: Optical Society of America, Jan. 14, 2008, 11 pages.

Defocus Video Matting, McGuire et al., Publisher: ACM SIGGRAPH, Jul. 31, 2005, 11 pages.

Compact multi-aperture imaging with high angular resolution, Santacana et al., Publisher: Optical Society of America, 2015, 10 pages.

Multi-Aperture Photography, Green et al., Publisher: Mitsubishi Electric Research Laboratories, Inc., Jul. 2007, 10 pages.

Multispectral Bilateral Video Fusion, Bennett et al., Publisher: IEEE, May 2007, 10 pages.

Super-resolution imaging using a camera array, Santacana et al., Publisher: Optical Society of America, 2014, 6 pages.

Optical Splitting Trees for High-Precision Monocular Imaging, McGuire et al., Publisher: IEEE, 2007, 11 pages.

High Performance Imaging Using Large Camera Arrays, Wilburn et al., Publisher: Association for Computing Machinery, Inc., 2005, 12 pages.

Real-time Edge-Aware Image Processing with the Bilateral Grid, Chen et al., Publisher: ACM SIGGRAPH, 2007, 9 pages.

Superimposed multi-resolution imaging, Carles et al., Publisher: Optical Society of America, 2017, 13 pages.

Viewfinder Alignment, Adams et al., Publisher: Eurographics, 2008, 10 pages.

Dual-Camera System for Multi-Level Activity Recognition, Bodor et al., Publisher: IEEE, Oct. 2014, 6 pages.

Engineered to the task: Why camera-phone cameras are different, Giles Humpston, Publisher: Solid State Technology, Jun. 2009, 3 pages.

* cited by examiner

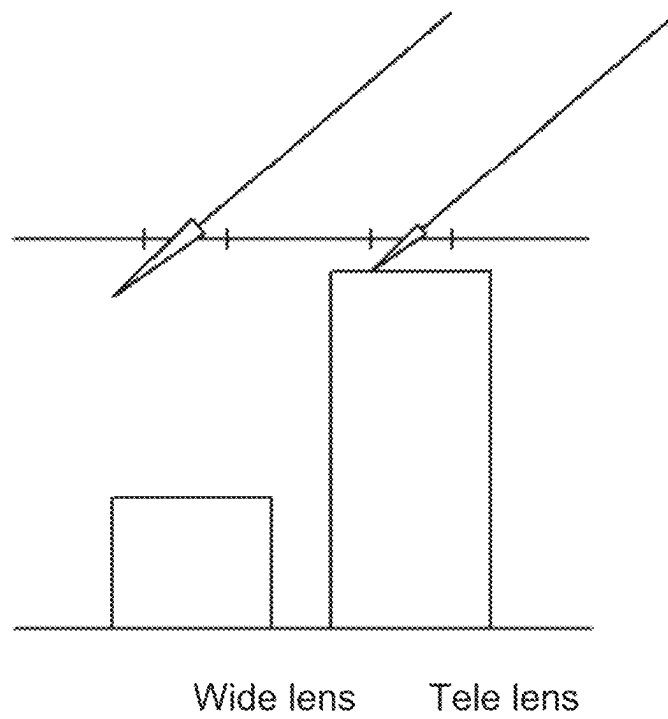


FIG. 1A

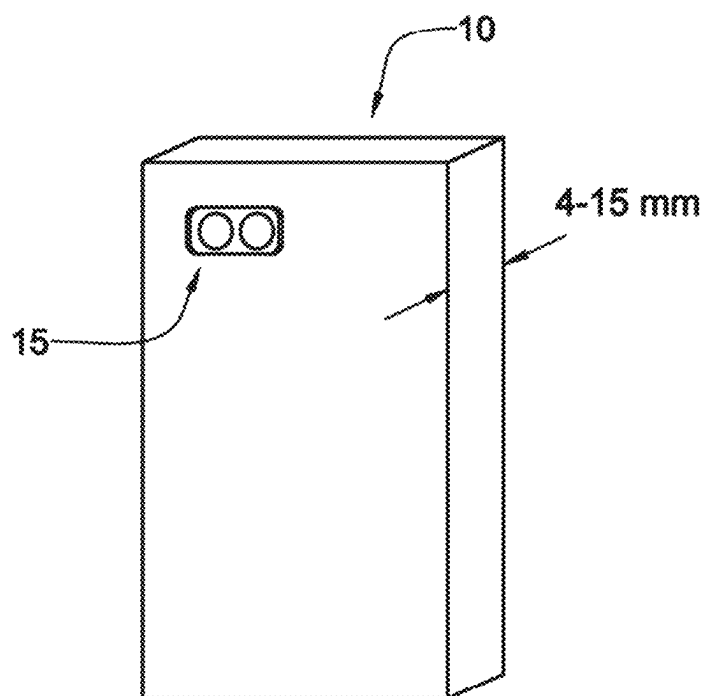


FIG. 1B

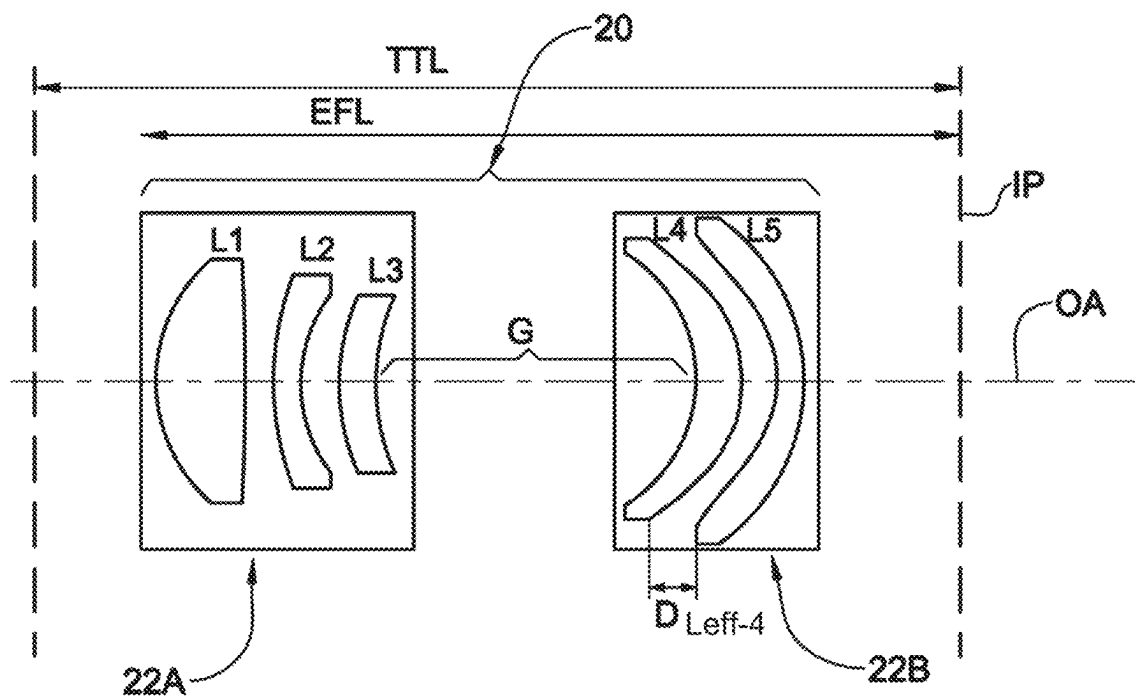


FIG. 1C

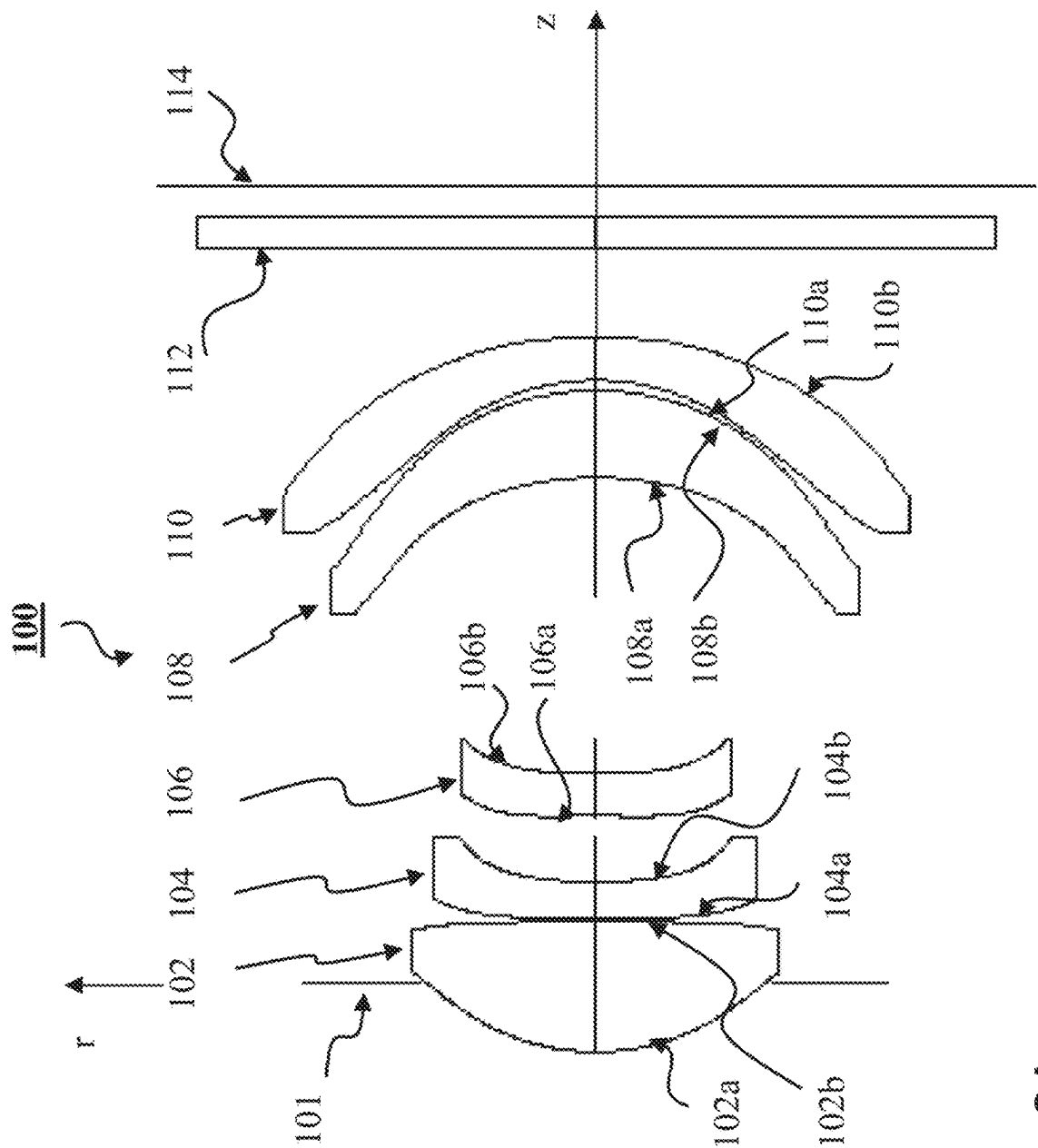


FIG. 2A

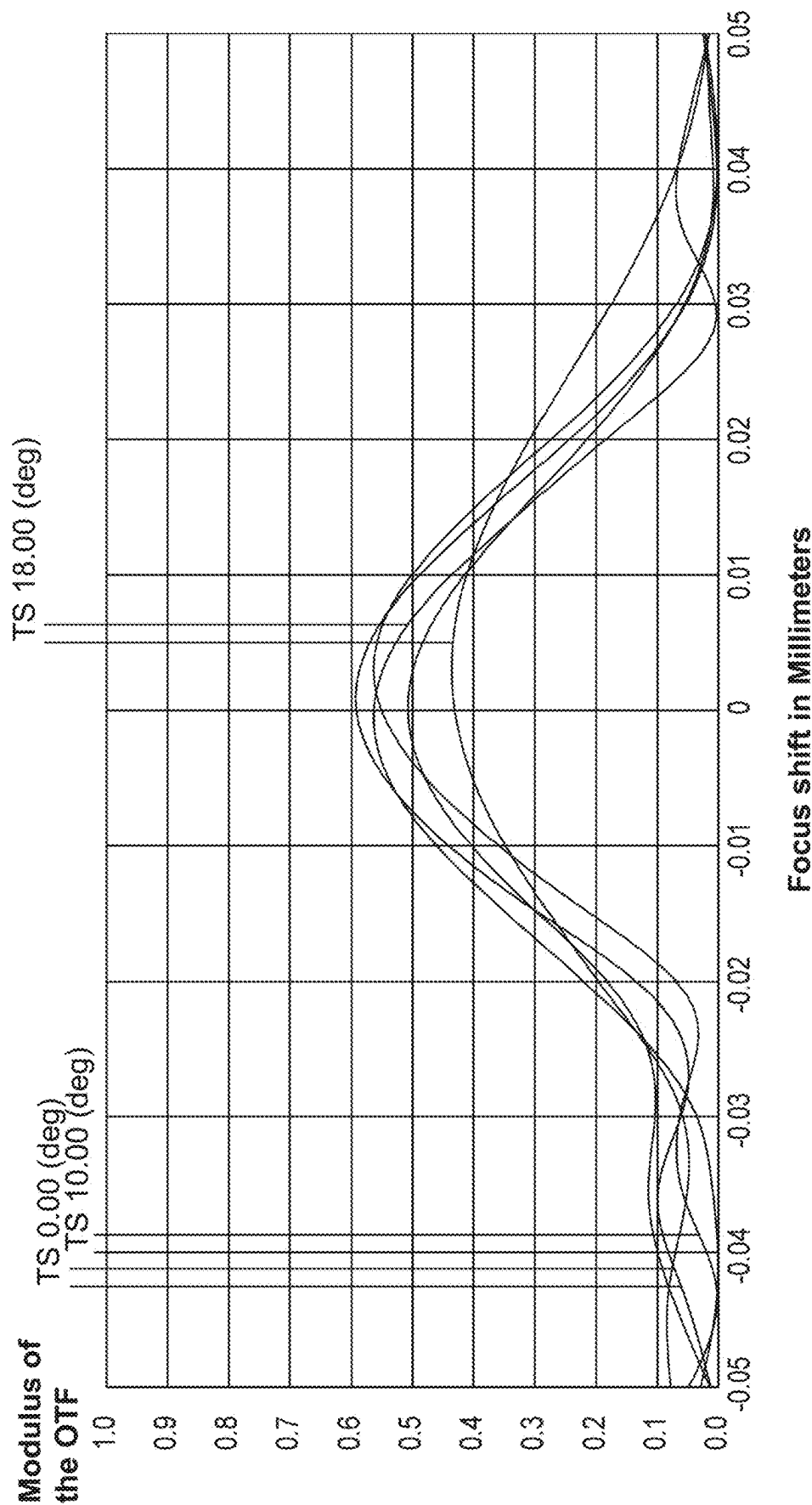
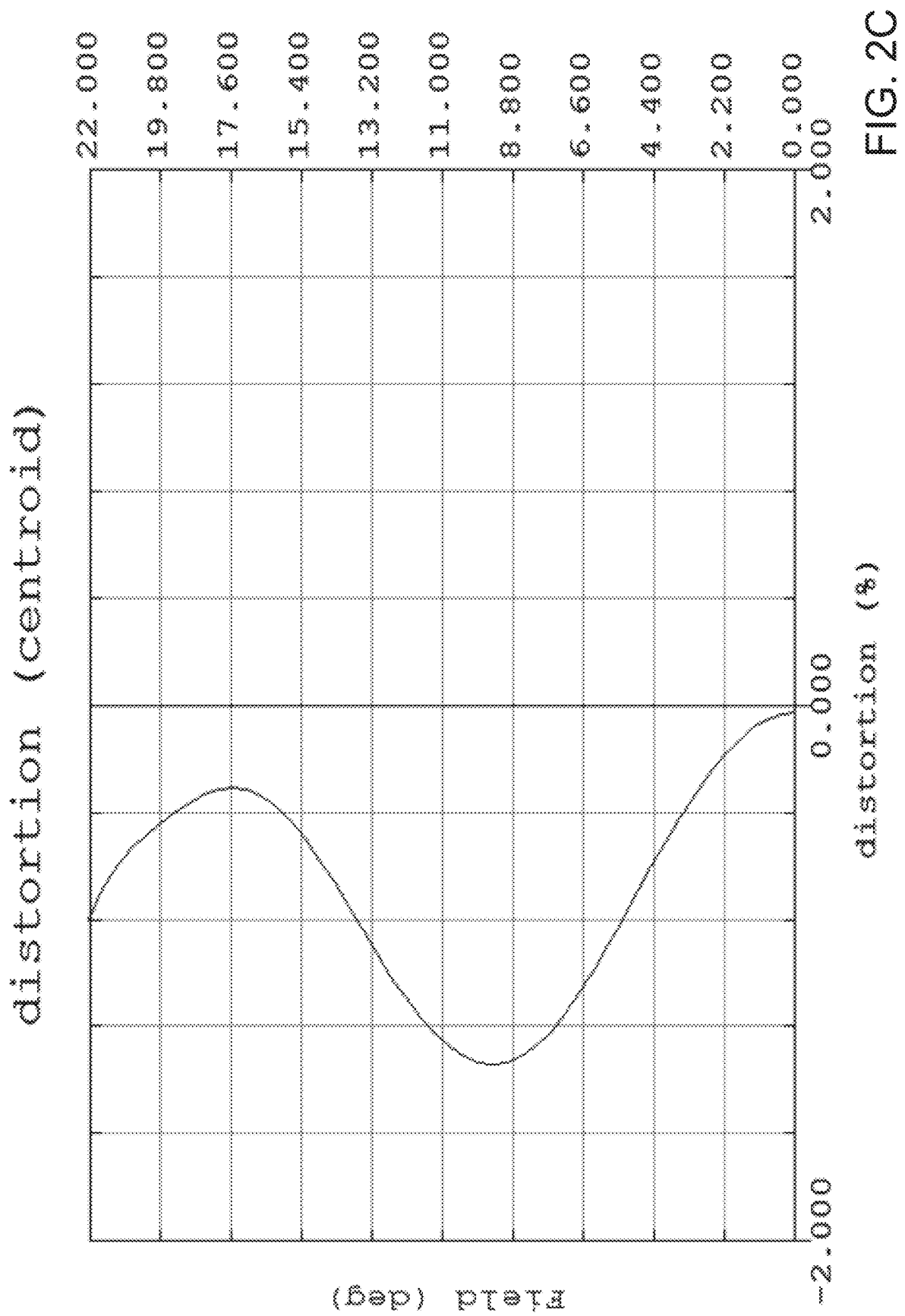


FIG. 2B



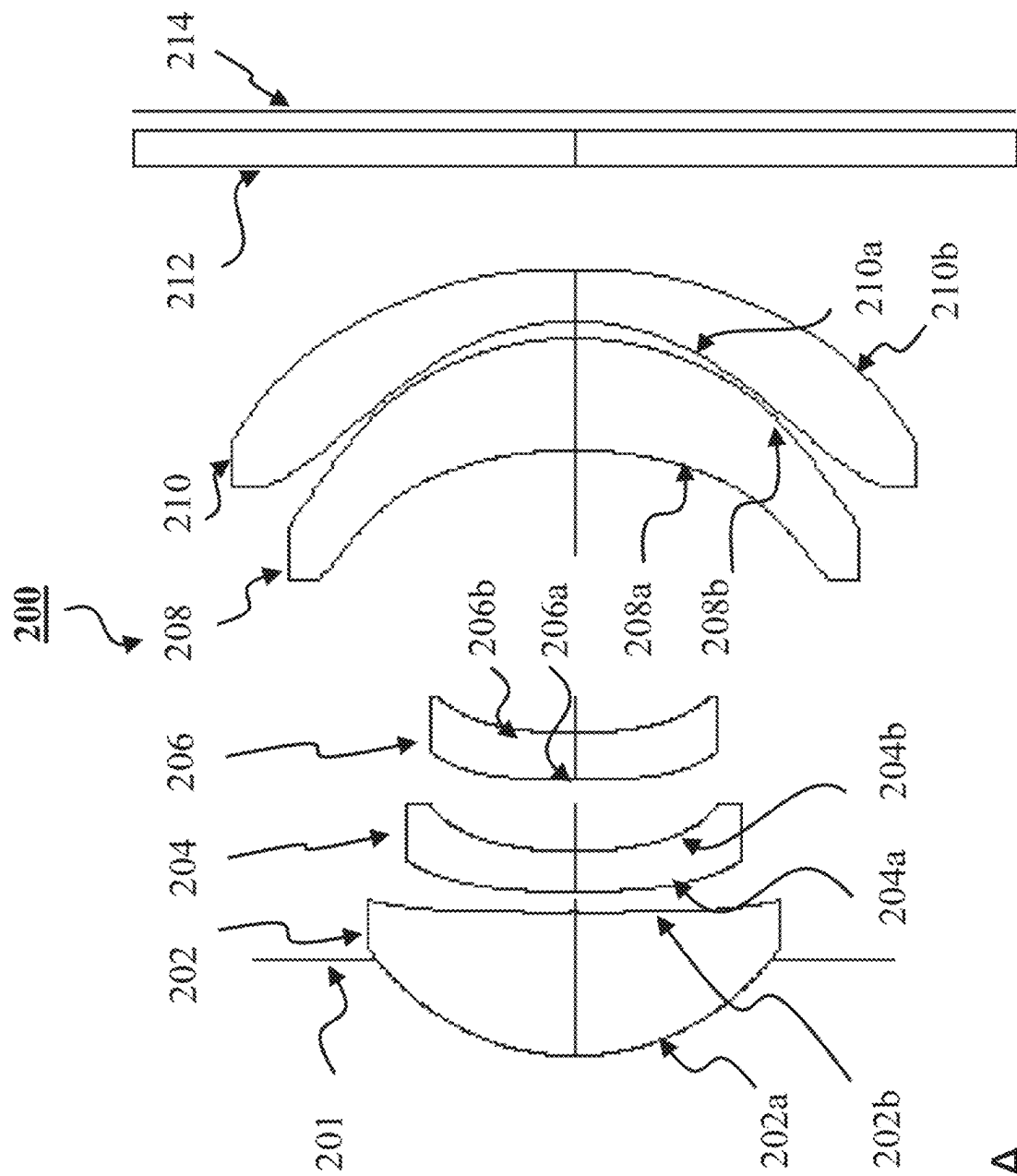


FIG. 3A

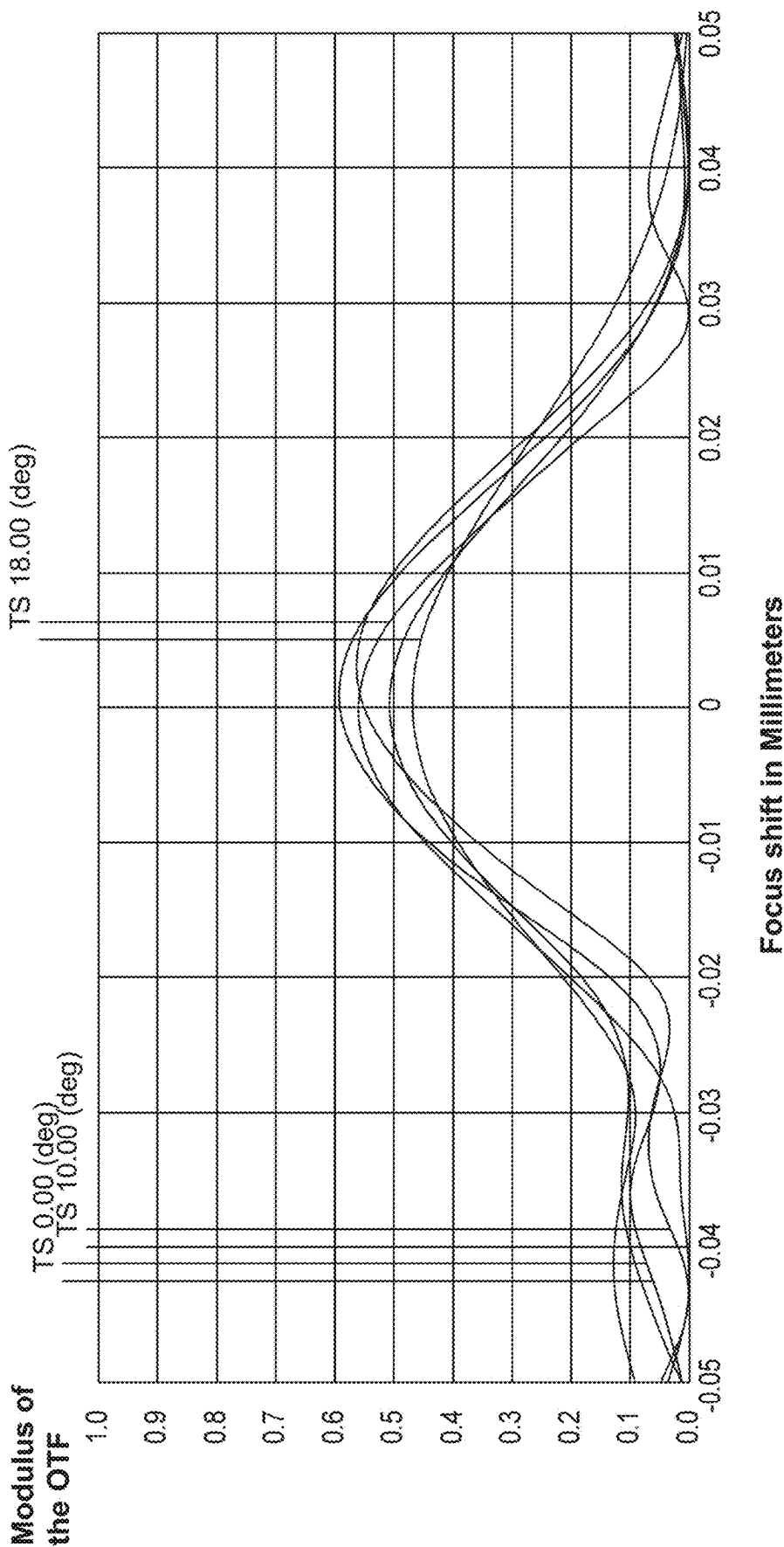


FIG. 3B

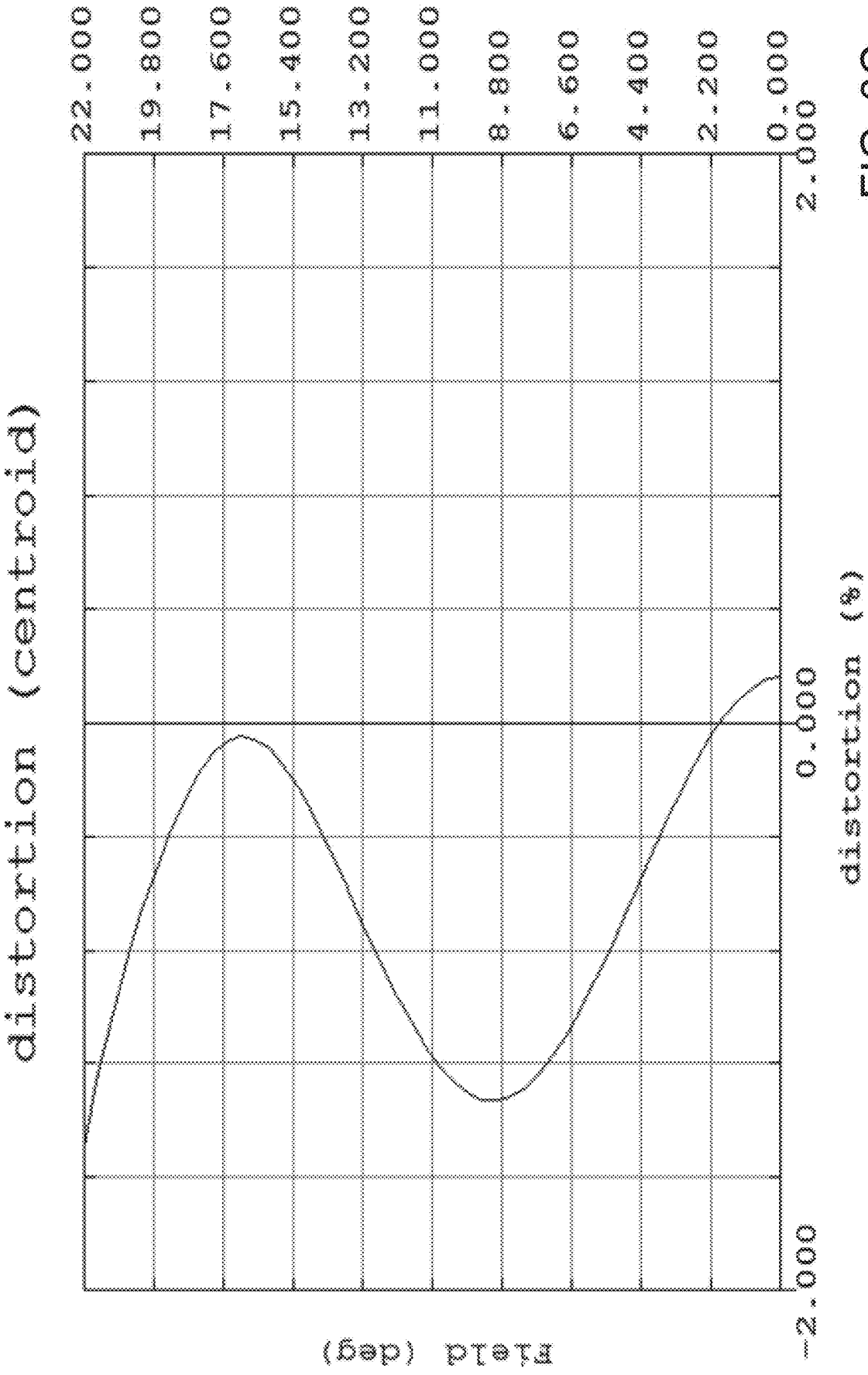


FIG. 3C

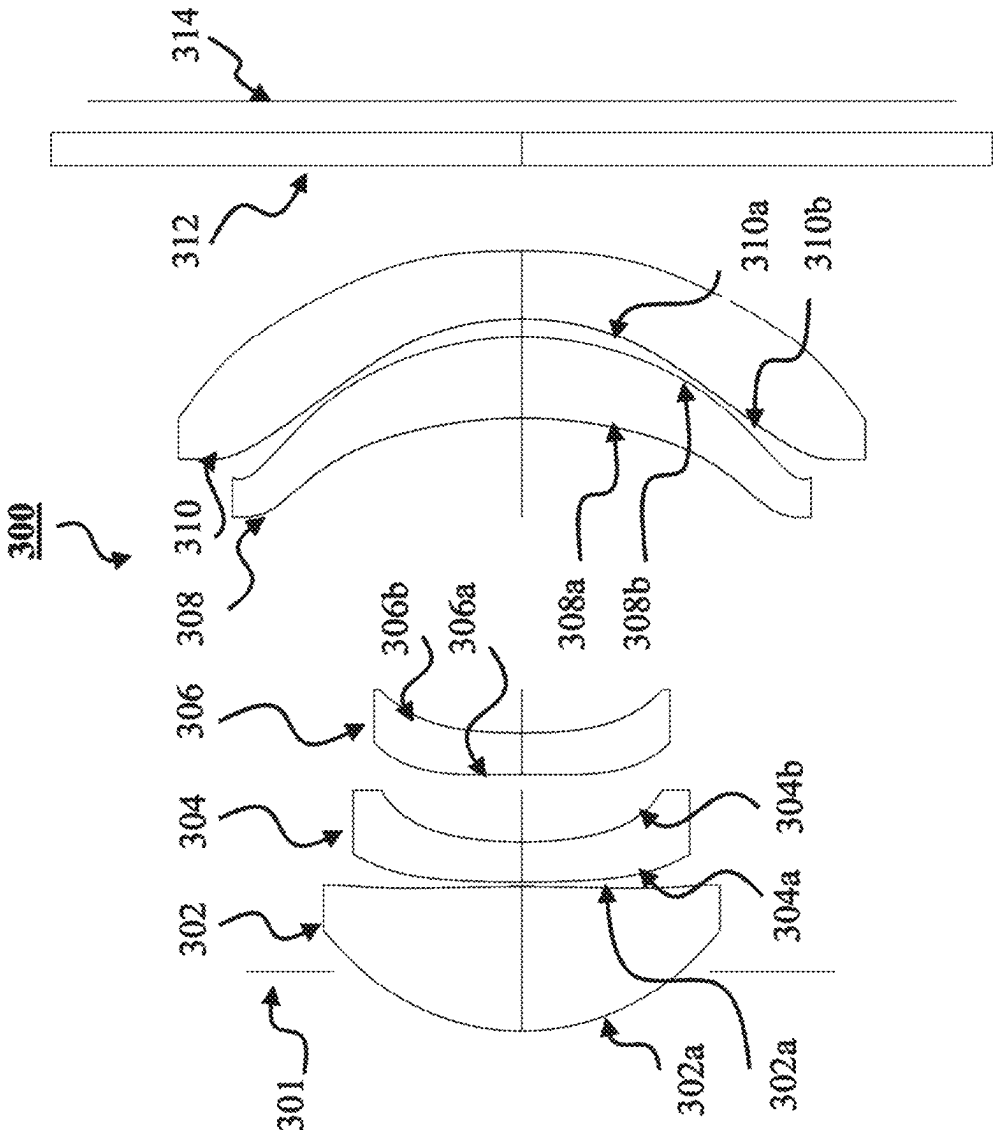


FIG. 4A

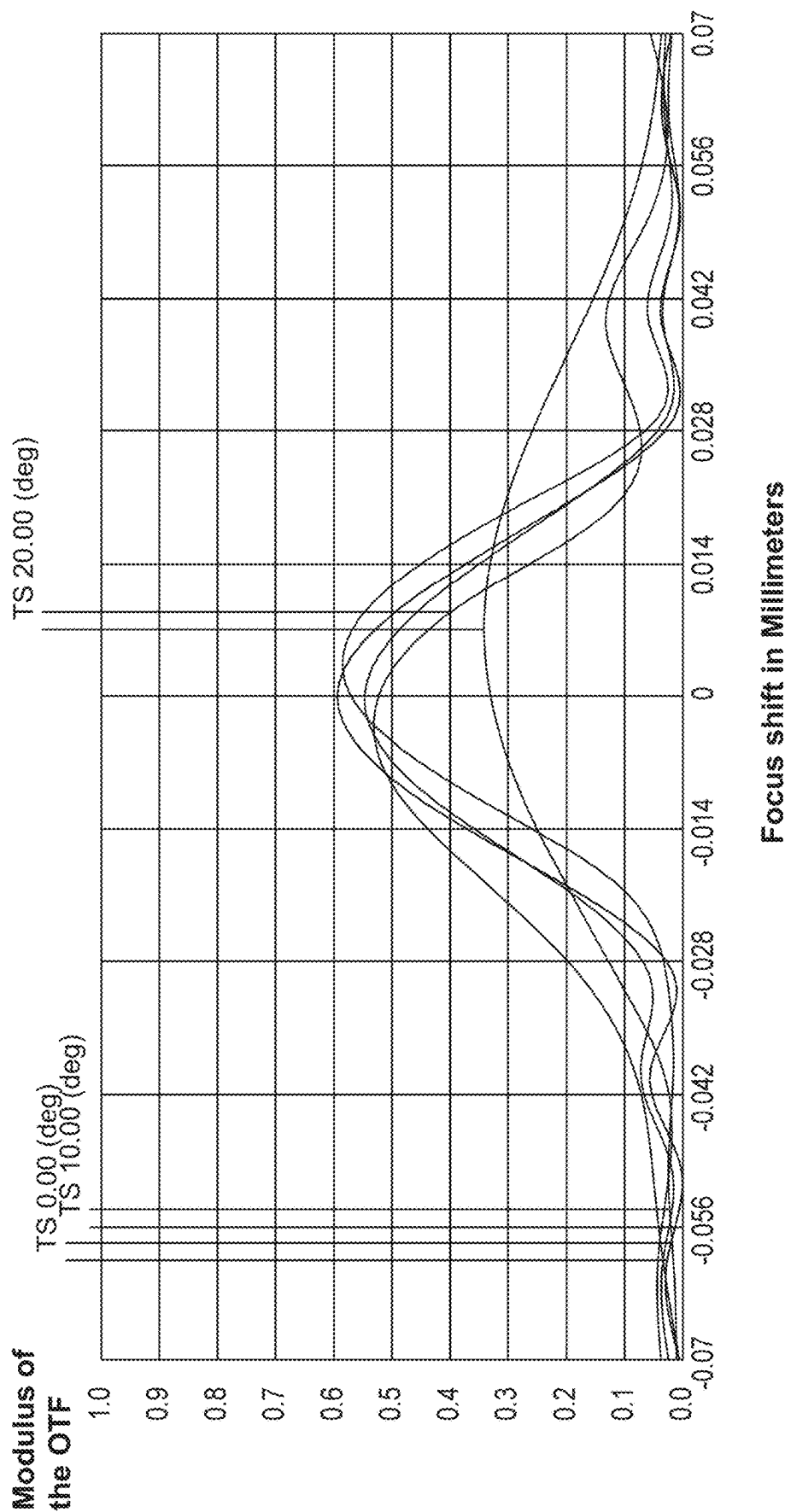


FIG. 4B

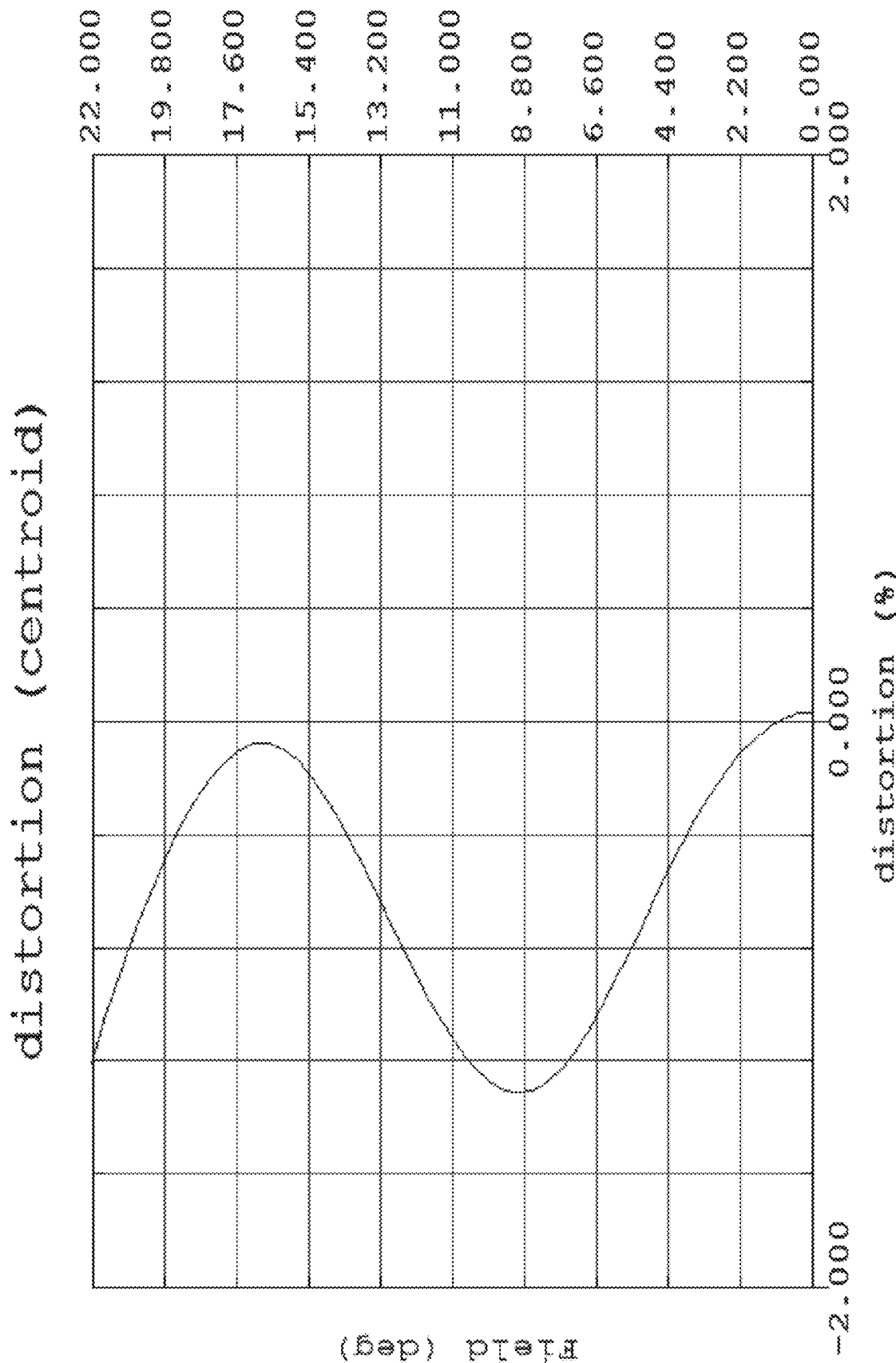
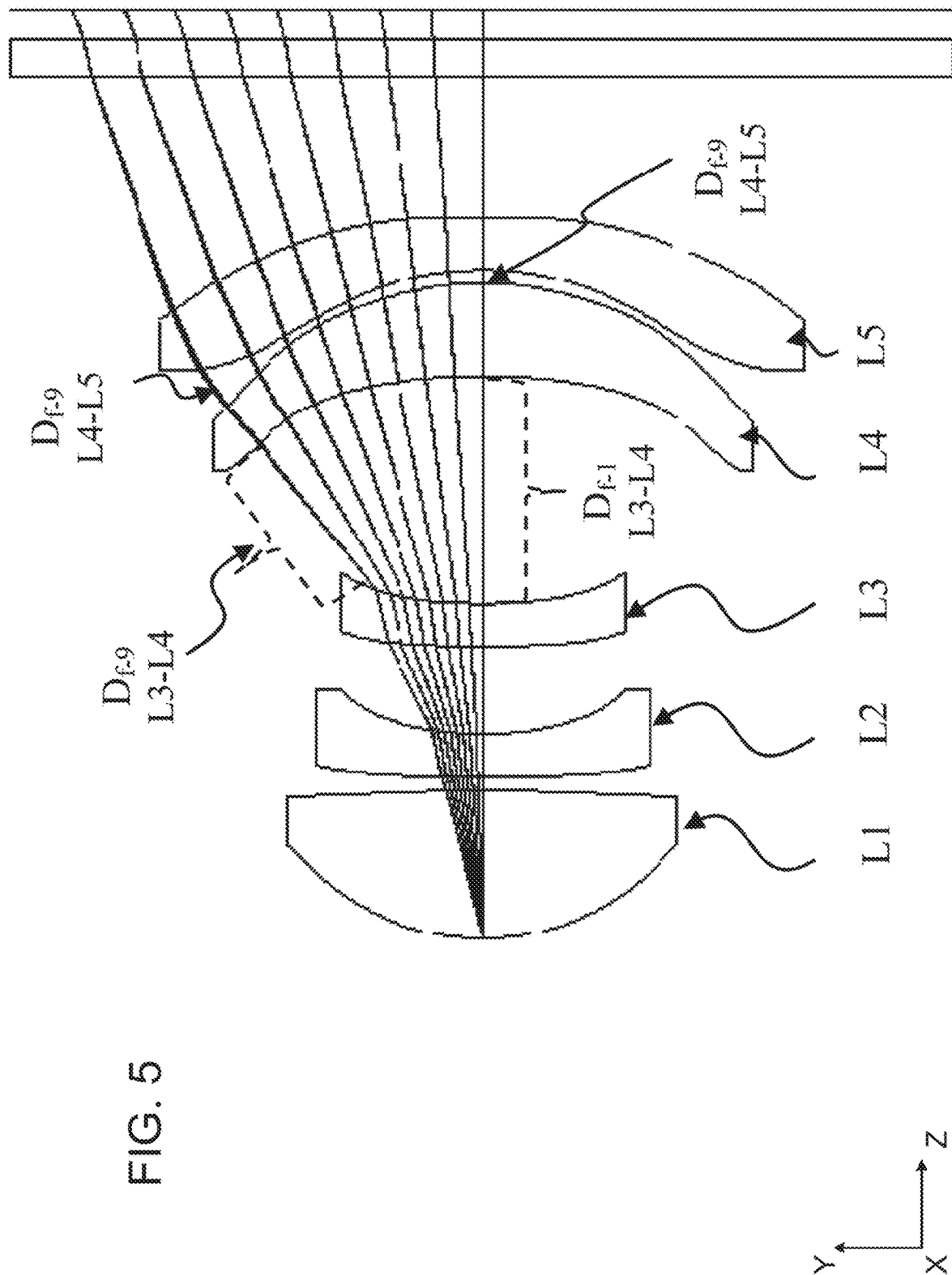
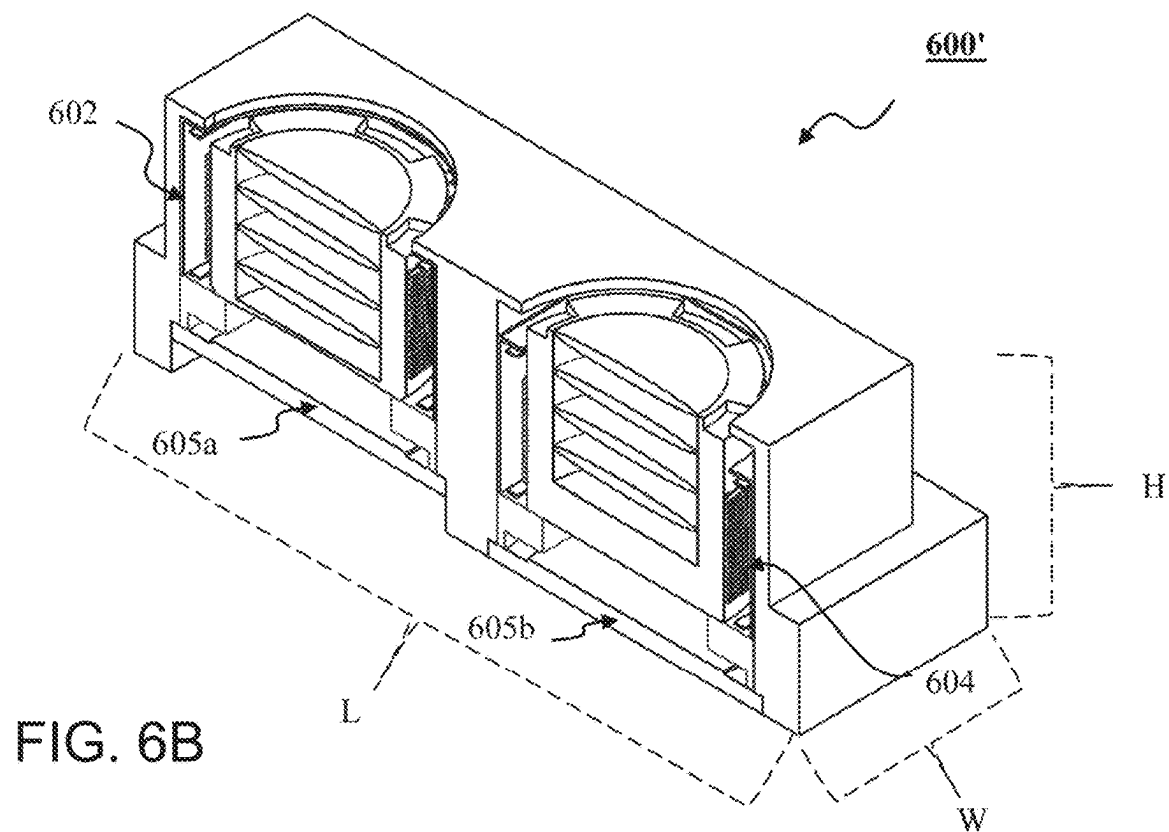
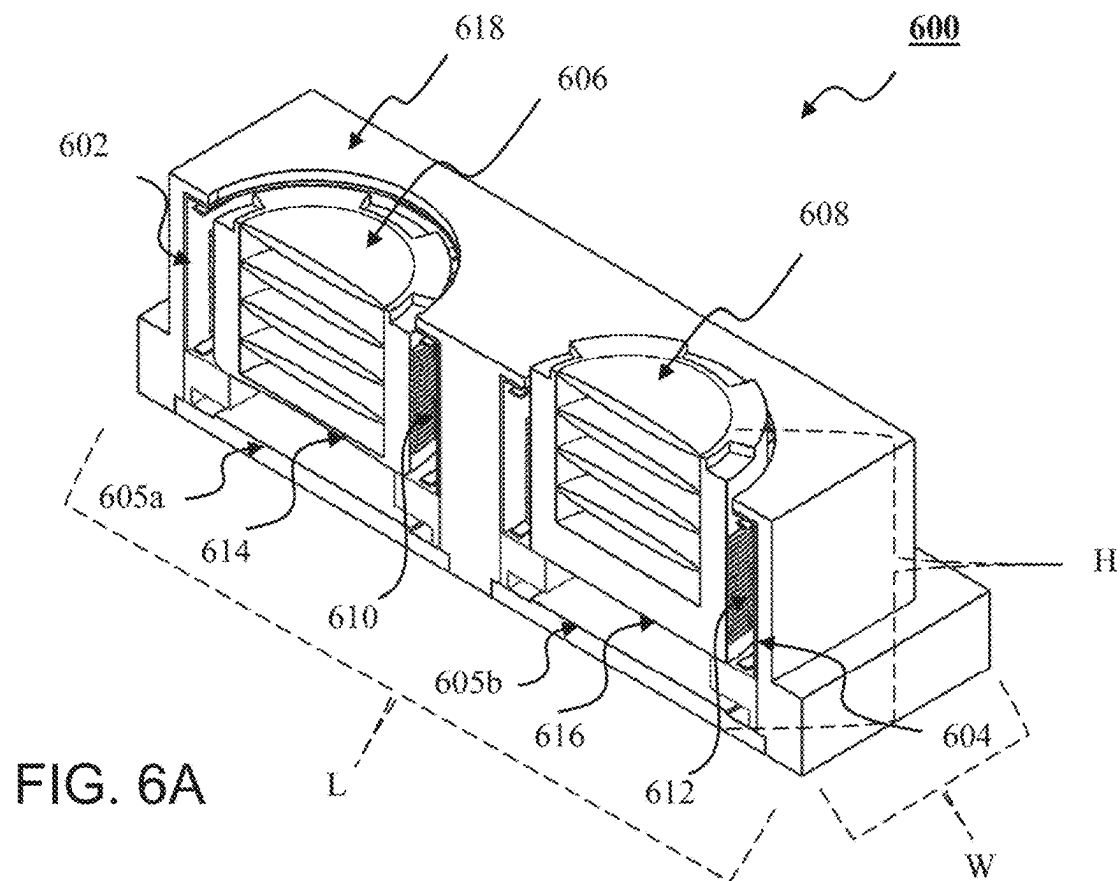
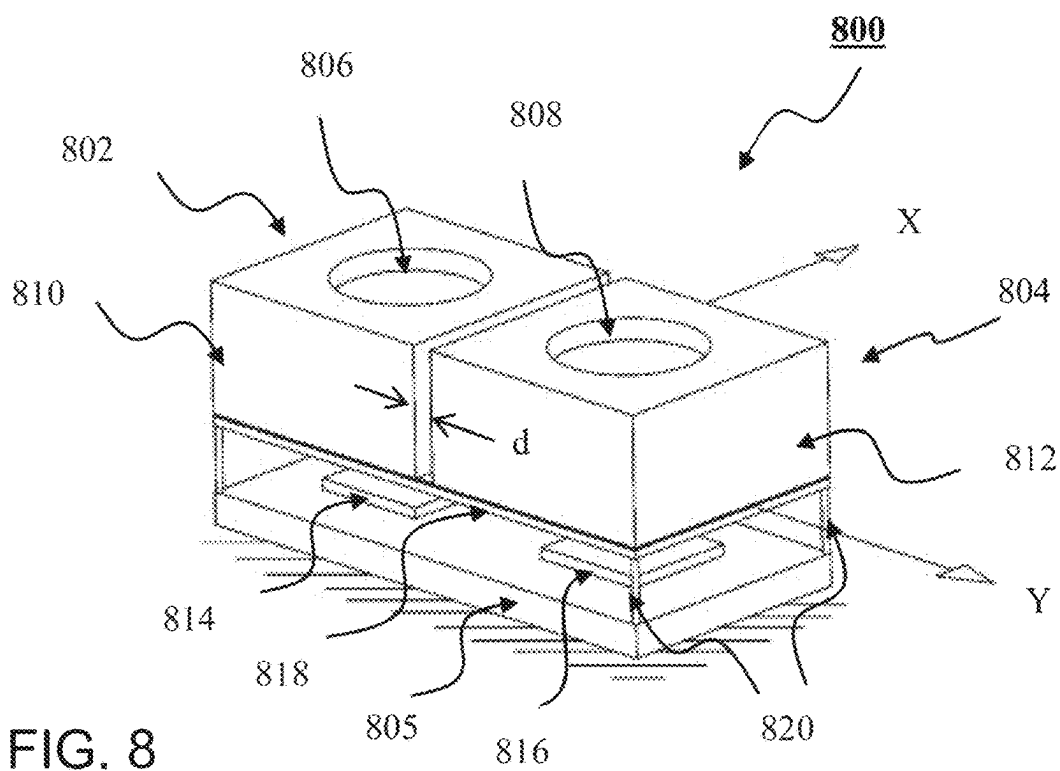
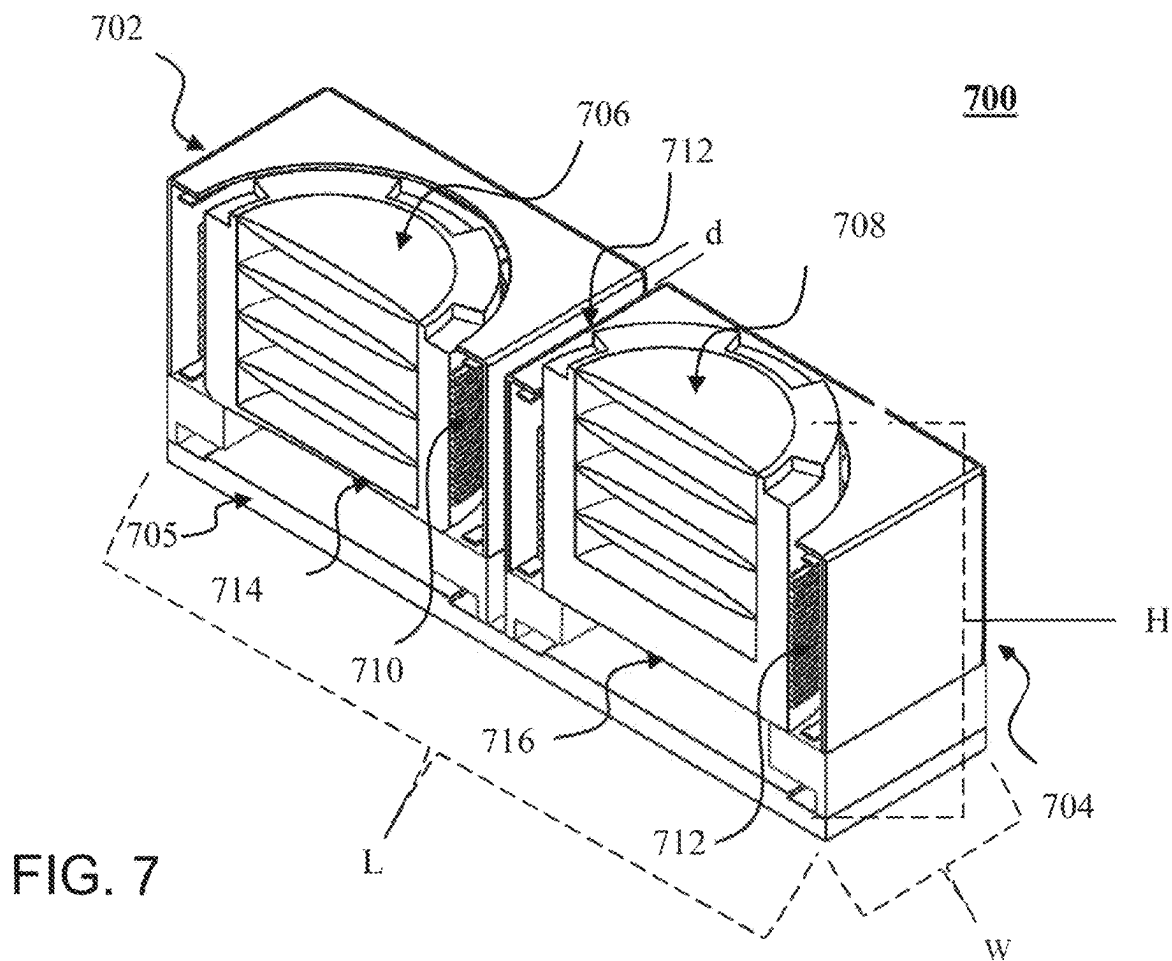


FIG. 4C



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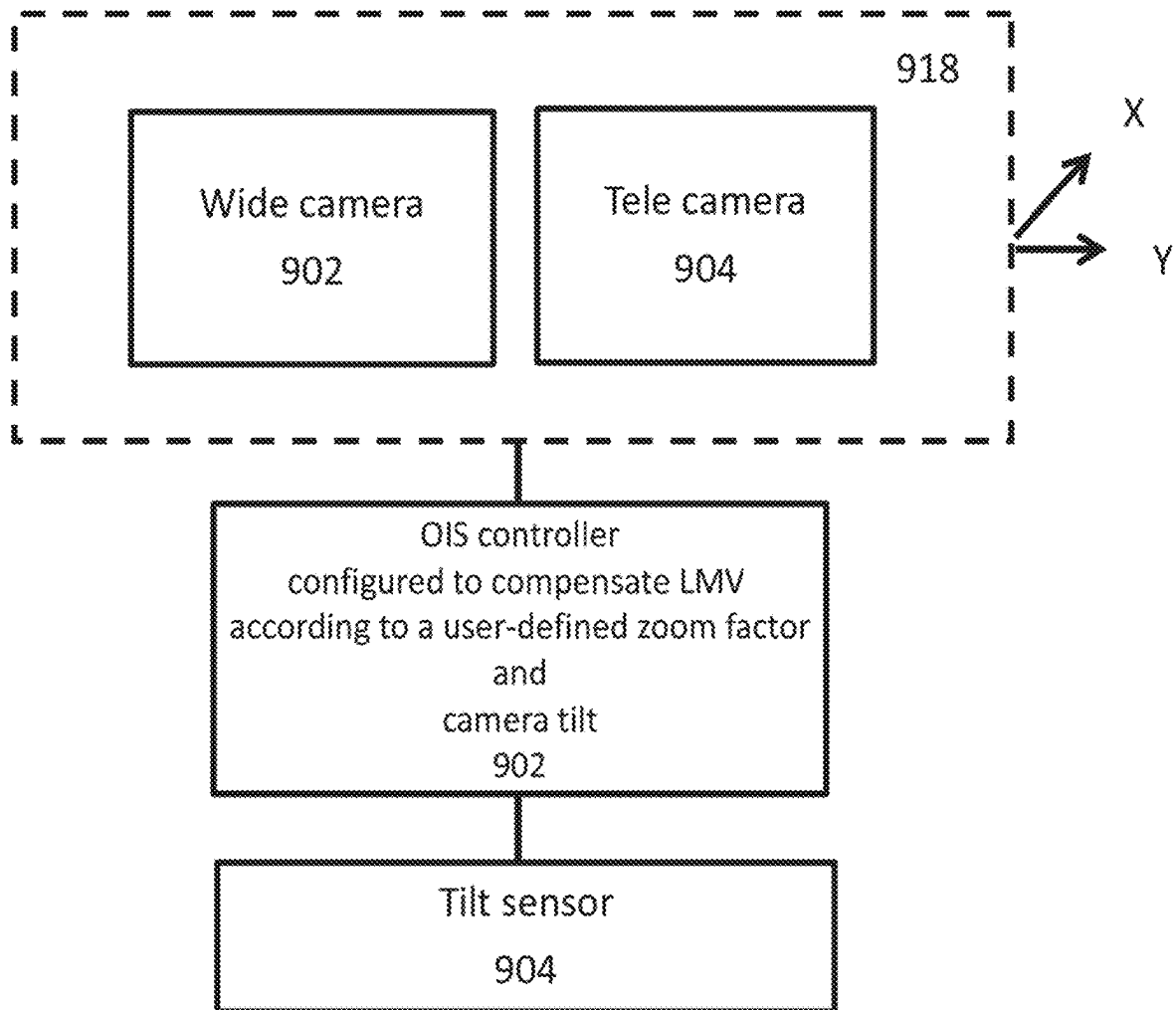


FIG. 9

MINIATURE TELEPHOTO LENS MODULE AND A CAMERA UTILIZING SUCH A LENS MODULE

CROSS REFERENCE TO EXISTING APPLICATIONS

This application is a Continuation application from U.S. patent application Ser. No. 18/641,478 filed Apr. 22, 2024 (now allowed), which was a Continuation application from U.S. patent application Ser. No. 17/389,510 filed Jul. 30, 2021 (now U.S. Pat. No. 11,994,654), which was a Continuation application from U.S. patent application Ser. No. 16/276,034 filed Feb. 14, 2019 (now U.S. Pat. No. 11,125,975), which was a Continuation application from U.S. patent application Ser. No. 15/540,676 filed Jun. 29, 2017 (now U.S. Pat. No. 10,288,840), which was a 371 from international patent application No. PCT/IB2015/050044 filed Jan. 3, 2015.

TECHNOLOGICAL FIELD

The present invention is generally in the field of imaging techniques, and relates to a camera and mobile electronic devices utilizing such a camera.

BACKGROUND

Digital camera modules are currently being incorporated into a variety of portable electronic devices. Such devices include for example mobile phones (e.g. smartphones), personal data assistants (PDAs), computers, and so forth. Digital camera modules for use in portable devices have to meet certain requirements such as good quality imaging, small footprint, as well as low weight.

Several techniques for small digital camera modules providing good quality imaging are described in WO14083489 and WO14199338, both assigned to the assignee of the present application.

According to the technique described in WO14083489, a multi-aperture imaging system comprises a first camera with a first sensor that captures a first image and a second camera with a second sensor that captures a second image. The two cameras have either identical or different FOVs. Either image may be chosen to be a primary or an auxiliary image, based on a zoom factor. An output image with a point of view determined by the primary image is obtained by registering the auxiliary image to the primary image.

The technique described in WO14199338 relates to a dual-aperture zoom digital camera operable in both still and video modes. The camera includes Wide and Tele imaging sections with respective lens/sensor combinations and image signal processors and a camera controller operatively coupled to the Wide and Tele imaging sections. The controller is configured to combine in still mode at least some of the Wide and Tele image data to provide a fused output image from a particular point of view, and to provide, without fusion, continuous zoom video mode output images, each output image having a given output resolution. The video mode output images are provided with a smooth transition when switching between a lower zoom factor (ZF) value and a higher ZF value or vice versa. At the lower ZF the output resolution is determined by the Wide sensor, while at the higher ZF value the output resolution is determined by the Tele sensor.

GENERAL DESCRIPTION

There is a need in the art for a novel camera module for use in modern portable electronic devices, such as smart phones, laptops, notepads, etc.

As noted above, the requirements for the camera modules for use in such devices are related to the size, weight and image quality of the camera. Moreover, these requirements become more essential when the camera module is to be installed within the portable device, unlike other external camera units attachable to the portable device. In the case of an internal (integral) camera unit, the dimensions of the camera optics should be as small as possible in order to be suitable to operate with commonly used detectors and to fit the thickness of the device in which the camera is installed (preferably without protruding from the device's casing), while the trend in such devices is to reduce the thickness as much as possible.

This problem is even more crucial when using, in a portable device, a lens with a long length with a fixed and relatively high zooming effect. Considering for example the dual-aperture zoom digital camera described in above-indicated publications WO14083489 and WO14199338 mentioned above, it utilizes Wide and Tele imaging channels which provide advanced imaging capabilities such as zoom and image quality by image fusion between the two channels.

One of the problems with dual-aperture zoom cameras relates to the dimensions (heights) of Wide and Tele cameras along the optical axis. Such dimensions depend on total track lengths (TTLs) of the Tele and Wide lenses used in the respective imaging channels.

As schematically illustrated in FIG. 1B, the TTL is typically defined as the maximal distance between the object-side surface of the lens module and an image plane IP defined by such a lens module (where the sensing surface of a camera detector is placed). In most miniature lenses, the TTL is larger than the effective focal length (EFL) of the lens module, which is equal to the distance between the effective principal plane of the lens and its focal plane (which substantially coincides with image plane IP).

With regard to the term effective principal plane, the following should be understood. Generally, the lens (or lens module) has front and rear principal planes, which have the property that a ray emerging from the lens appears to have crossed the rear principal plane at the same distance from the axis that that ray appeared to cross the front principal plane, as viewed from the front of the lens. This means that the lens can be treated as if all of the refraction occurred at the principal planes. The principal planes are crucial in defining the optical properties of the system, since it is the distance of the object and image from the front and rear principal planes that determine the magnification of the system. The principal points are the points where the principal planes cross the optical axis.

Considering dual-aperture optical zoom in a mobile phone (e.g. a smartphone) with the typically used lenses, i.e. typical TTL/EFL ratio of about 1.3, the Wide and Tele lenses would have TTLs of about 4.55 mm and 9.1 mm, respectively. This will result in undesirably long camera modules for use in such a smartphone device.

Further, the difference in the TTLs of the Wide and Tele lens modules can cause shadowing and light-blocking problems. Reference is made to FIG. 1A schematically illustrating that part of incoming light incident on the "higher" lens does not reach the "shorter" lens. In this connection, one should keep in mind that a distance between the Tele and

Wide lens modules should be as small as possible to meet the overlapping/common FOVs as well as footprint requirements for the camera unit in a portable device.

Another part of the presently disclosed subject matter is associated with the implementation of standard optical image stabilization (OIS) in a dual-aperture zoom camera. Standard OIS compensates for camera tilt ("CT") by a parallel-to-the image sensor (exemplarily in the X-Y plane) lens movement ("LMV"). Camera tilt causes image blur. The amount of LMV (in mm) needed to counter a given camera tilt depends on the camera's lens EFL, according to the relation $LMV = CT * EFL$ where "CT" is in radians and EFL is in mm. Since, as shown above, a dual-aperture zoom camera may include two lenses with significantly different EFLs, it is impossible to move both lenses together and achieve optimal tilt compensation for both Tele and Wide cameras. That is, since the tilt is the same for both cameras, a movement that will cancel the tilt for the Wide camera will be insufficient to cancel the tilt for the Tele camera. Similarly, a movement that will cancel the tilt for the Tele camera will over-compensate the tilt cancellation for the Wide camera. Assigning a separate OIS actuator to each camera can achieve simultaneous tilt compensation, but at the expense of a complicated and costly camera system.

Thus, for both a single-aperture or multi-aperture (dual) camera unit, the use of a telephoto lens would be advantageous, as such a telephoto lens provides reduced TTL while enabling to maintain the relatively high EFL required for the Tele lens, i.e. for telephoto lens $TTL < EFL$. However, the dimensions of conventional lenses in which the telephoto condition is satisfied do not allow them to be used as integral lenses fully embedded in a thin portable device. The telephoto lens module, in order to be used as an integral lens in a modern portable device, has to satisfy the telephoto condition (i.e. $TTL < EFL$) while the lens module is to be as short as possible (along the optical path of light passing through it) allowing it to be fully fitted within the portable device casing.

Accordingly, a miniature telephoto lens module is disclosed which is designed with the desired dimensions to enable its integration within a portable device. According to some examples of the presently disclosed subject matter, the miniature telephoto lens module (or telephoto lens unit) is designed to be completely integrated within the casing of a conventional Smartphone, i.e. without protruding therefrom. The disclosed telephoto lens module has a total track lens (TTL) smaller than an effective focal lens (EFL) thereof, and is configured such that its dimension along the optical axis is desirably small, i.e. about 4-15 mm or less (e.g. suitable to be fitted in a portable device having a casing as small as 4 mm).

The telephoto lens unit comprises multiple lens elements made of at least two different polymer materials having different Abbe numbers. The multiple lens elements comprise a first group of at least three lens elements being a telephoto lens assembly, and a second group of at least two lens elements being a field lens assembly.

The first group of lens elements comprises, in order from the object plane to the image plane along an optical axis of the telephoto lens unit: a first lens having positive optical power and a pair of second and third lenses having together negative optical power such that said telephoto lens assembly provides a telephoto optical effect of said telephoto lens unit and wherein said second and third lenses are each made of one of said at least two different polymer materials having a different Abbe number, for reducing chromatic aberrations of said telephoto lens. The second group of lens elements is

configured to correct field curvature of said telephoto lens assembly, and said field lens module comprises two or more of said lens elements made of the different polymer materials respectively having different Abbe numbers, and configured to compensate for residual chromatic aberrations of said telephoto lens assembly dispersed during light passage through an effective gap located between the telephoto and field lens assemblies. The effective gap is larger than $\frac{1}{5}$ of the TTL of the telephoto lens unit, thereby allowing sufficient field separation for reducing chromatic aberration.

Various examples disclosed herein include an optical lens unit comprising, in order from an object side to an image side: a first lens element with positive refractive power having a convex object-side surface, a second lens element with negative refractive power having a thickness d_2 on an optical axis and separated from the first lens element by a first air gap, a third lens element with negative refractive power and separated from the second lens element by a second air gap, a fourth lens element having a positive refractive power and separated from the third lens element by an effective third air gap, and a fifth lens element having negative refractive power, separated from the fourth lens element by an effective fourth air gap, the fifth lens element having a thickness d_5 on the optical axis.

An optical lens unit may further include a stop, positioned before the first lens element, a glass window disposed between the image-side surface of the fifth lens element and an image sensor with an image plane on which an image of the object is formed.

Each lens element has two surfaces, each surface having a respective diameter. The largest diameter among all lens elements is defined as an "optical diameter" of the lens assembly.

As disclosed herein, TTL is defined as the distance on an optical axis between the object-side surface of the first lens element and an image plane where the image sensor is placed. "EFL" has its regular meaning, as mentioned above. In all embodiments, TTL is smaller than the EFL, i.e. the TTL/EFL ratio is smaller than 1.0. In some embodiments, the TTL/EFL ratio is smaller than 0.9. In an embodiment, the TTL/EFL ratio is about 0.85. According to some examples the lens assembly has an F number $F \# < 3.2$.

According to an example disclosed herein, the focal length of the first lens element f_1 is smaller than $TTL/2$, the first, third and fifth lens elements have each an Abbe number ("Vd") greater than 50, the second and fourth lens elements have each an Abbe number smaller than 30, the first air gap is smaller than $d_2/2$, the effective third air gap is greater than $TTL/5$ and the effective fourth air gap is smaller than $1.5d_5TTL/50$. In some embodiments, the surfaces of the lens elements may be aspheric.

In the optical lens unit mentioned above, the first lens element with positive refractive power allows the TTL of the lens unit to be favorably reduced. The combined design of the first, second and third lens elements plus the relative short distances between them enable a long EFL and a short TTL. The same combination, together with the high dispersion (low Vd) for the second lens element and low dispersion (high Vd) for the first and third lens elements, also helps to reduce chromatic aberration. In particular, the ratio $TTL/EFL < 1.0$ and minimal chromatic aberration are obtained by fulfilling the relationship $1.2 \times |f_3| > |f_2| > 1.5 \times f_1$, where "f" indicates the lens element effective focal length and the numerals 1, 2, 3, 4, 5 indicate the lens element number.

The relatively large effective gap between the third and the fourth lens elements plus the combined design of the fourth and fifth lens elements assist in bringing all fields'

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focal points to the image plane. Also, because the fourth and fifth lens elements have different dispersions and have respectively positive and negative power, they help in minimizing chromatic aberration.

The telephoto lens module disclosed herein may be advantageously adapted to be incorporated in a mobile phone camera that uses a typical $\frac{1}{4}$ " or $\frac{1}{3}$ " image sensor. For example, to be competitive with known mobile phone cameras with $\frac{1}{4}$ " image sensors, it would be advantageous for the TTL of the telephoto lens module to be smaller than 5.5 mm and the largest lens diameter to be smaller than 4 mm. To be competitive with known mobile phone cameras with $\frac{1}{3}$ " image sensors, it would be advantageous for the TTL of the telephoto lens module to be smaller than 6.5 mm and the largest lens diameter to be smaller than 5 mm.

Accordingly to an example of the presently disclosed subject matter there is provided an optical lens unit configured to provide an image on an entire area of a $\frac{1}{4}$ " image sensor, the lens unit comprising five lens elements and having a TTL smaller than 5.5 mm, an EFL larger than 5.9 mm, and an optical diameter smaller than 4 mm.

Accordingly in another example of the presently disclosed subject matter there is provided an optical lens unit operative to provide an image on an entire area of a $\frac{1}{3}$ " image sensor, the lens unit comprising five lens elements and having a TTL smaller than 6.2 mm, an EFL larger than 6.8 mm, and an optical diameter smaller than 5 mm.

Also, as mentioned above, according to the presently disclosed subject matter it is suggested to have all lens elements made of polymer material such as plastic. While lenses made of polymer material are advantageous for reducing the price tag of the telephoto lens module as well as its weight, there are very few polymer materials which are suitable for this purpose. This is different to glass lenses which can be made of a variety of different glass materials, each characterized by a different Abbe number. The scarcity in polymer materials presents a challenge when designing lenses for a telephoto lens module. This challenge is at least partly due to the limitation in possible combinations of different lenses with different Abbe numbers which can be used for the purpose of correcting field curvature and compensating for chromatic aberrations.

Thus, according to one aspect of the presently disclosed subject matter there is provided a mobile electronic device comprising an integrated camera, wherein the camera comprises a Wide camera unit comprising a Wide lens unit, and a Telephoto camera unit comprising a telephoto lens unit, the telephoto lens unit and the wide lens unit having respectively TTL/EFL ratios smaller and larger than 1 and defining separate telephoto and wide optical paths.

In addition to the above features, the mobile electronic device according to this aspect of the presently disclosed subject matter can optionally comprise one or more of features (i) to (xvi) below, in any desired combination or permutation:

- (i). wherein light receiving outer surfaces of the Wide and Telephoto lens units are located substantially in the same plane, thereby reducing shadowing and light blocking effects therebetween.
- (ii). wherein the Wide and Telephoto camera units are mounted on separate printed circuit boards.
- (iii). wherein the printed circuit boards are located in different spaced-apart substantially parallel planes.
- (iv). wherein the Wide and Telephoto camera units are mounted directly on a single printed circuit board.
- (v). wherein the Wide and Telephoto camera units are spaced from one another a distance d of about 1 mm.

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(vi). wherein the telephoto lens unit is made of at least two polymer materials.

(vii). wherein the telephoto lens has a total track lens (TTL) not exceeding 15 mm.

(viii). wherein the telephoto lens has TTL less than 10 mm.

(ix). wherein the telephoto lens unit comprises multiple lens elements made of at least two different polymer materials having different Abbe numbers, the multiple lens elements comprise a first group of at least three lens elements configured to form a telephoto lens assembly, and a second group of at least two lens elements configured to form a field lens assembly, wherein the field lens assembly is spaced from the telephoto lens assembly by a predetermined effective gap.

(x). wherein said at least two different polymer materials comprise at least one plastic material with the Abbe number larger than 50, and at least one plastic material with the Abbe number smaller than 30.

(xi). wherein the first group of lens elements comprises, in order from an object plane to an image plane along an optical axis of the telephoto lens unit: a first lens having positive optical power and a pair of second and third lenses having together negative optical power such that said telephoto lens assembly provides telephoto optical effect of said telephoto lens unit, and said second and third lenses are each made of one of said at least two different polymer materials having a different Abbe number, for reducing chromatic aberrations of said telephoto lens; and

the second group of lens elements is configured to correct field curvature of said telephoto lens assembly, and comprises two or more of said lens elements made of the different polymer materials respectively having different Abbe numbers, and configured to compensate for residual chromatic aberrations of said telephoto lens assembly dispersed during light passage through said effective gap between the telephoto and field lens assemblies.

(xii). wherein the first, third and fifth lens elements have each an Abbe number greater than 50, and the second and fourth lens elements have each an Abbe number smaller than 30.

(xiii). wherein the predetermined effective gap is equal to or larger than $\frac{1}{5}$ of the TTL of the telephoto lens unit.

(xiv). wherein the lens elements of the field lens assembly are spaced from one another an effective air gap smaller than $\frac{1}{50}$ of the TTL of the telephoto lens unit.

(xv). wherein the telephoto lens unit has a TTL smaller than 5.5 mm, an effective focal length (EFL) larger than 5.9 mm, and an optical diameter smaller than 4 mm, thereby enabling to provide an image on an entire area of a $\frac{1}{4}$ " image sensor.

(xvi). wherein the telephoto lens unit has a TTL smaller than 6.2 mm, an effective focal length (EFL) larger than 6.8 mm, and an optical diameter smaller than 5 mm, thereby enabling to provide an image on an entire area of a $\frac{1}{3}$ " image sensor.

According to another aspect of the presently disclosed subject matter there is provided a camera for integrating in a mobile electronic device, the camera comprising a Wide camera unit and a Telephoto camera unit comprising respectively a wide lens unit and a telephoto lens unit having TTL/EFL ratios larger and smaller than 1, respectively, and defining wide and telephoto optical paths.

Wherein according to some examples the lens elements of at least the telephoto lens unit are made of one or more polymer materials.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to better understand the subject matter that is disclosed herein and to exemplify how it may be carried out in practice, embodiments will now be described, by way of non-limiting example only, with reference to the accompanying drawings, in which:

FIG. 1A is a schematic illustration demonstrating shadowing and light-blocking problems caused by height differences between Wide and Tele cameras in a dual-aperture camera;

FIG. 1B is a schematic illustration of a mobile phone device (constituting a portable electronic device) utilizing a camera unit as disclosed herein which is fully integrated inside the smartphone device;

FIG. 1C is a schematic illustration of a telephoto lens unit according to the presently disclosed subject matter;

FIG. 2A is a schematic illustration of a specific configuration of the telephoto lens unit, according to a first example of the presently disclosed subject matter;

FIG. 2B shows a graph plotting the modulus of the optical transfer function (MTF) vs. focus shift of the entire optical lens unit of FIG. 2A for various fields;

FIG. 2C shows a graph plotting the distortion vs. field angle (+Y direction) for the lens unit of FIG. 2A;

FIG. 3A is a schematic illustration of another possible configuration of the telephoto lens unit, according to a first example of the presently disclosed subject matter;

FIG. 3B shows a graph plotting the MTF vs. focus shift of the entire optical lens assembly for various fields in the lens unit of FIG. 3B, according to the second example of the presently disclosed subject matter;

FIG. 3C shows a graph plotting the distortion +Y in percent for the lens unit of FIG. 3A;

FIG. 4A is a schematic illustration of a specific configuration of the telephoto lens unit, according to a first example of the presently disclosed subject matter;

FIG. 4B shows a graph plotting the MTF vs. focus shift of the entire optical lens system for various fields in the lens unit of FIG. 4A;

FIG. 4C shows a graph plotting the distortion +Y in percent for the lens unit of FIG. 4A;

FIG. 5 is a schematic illustration showing the concept of an effective air gap between adjacent lenses in an optical lens unit, according to the presently disclosed subject matter;

FIG. 6A is a schematic illustration, in perspective cross section, of an example of a dual-aperture zoom camera, with each camera on a separate printed circuit board (PCB), according to the presently disclosed subject matter;

FIG. 6B is a schematic illustration, in perspective cross section, of another example of a dual-aperture zoom camera, with each camera on a separate PCB, according to the presently disclosed subject matter;

FIG. 7 is a schematic illustration, in perspective cross section, of yet another example of a dual-aperture zoom camera, where both cameras are mounted on a single PCB, according to the presently disclosed subject matter;

FIG. 8 is a schematic illustration of an example of a dual-aperture zoom camera that includes an OIS mechanism, according to the presently disclosed subject matter; and

FIG. 9 shows schematically a functional block diagram of the camera example of FIG. 8, according to the presently disclosed subject matter.

DETAILED DESCRIPTION OF EMBODIMENTS

The present invention includes novel configuration of a lens unit in a portable camera, advantageously applicable in a portable electronic device. This is schematically illustrated in FIG. 1B. In this example, such a portable electronic device **10** is constituted by a mobile phone device (e.g. smartphone). The mobile device is typically a few millimeters thick, e.g. 4 mm-15 mm.

However, as explained above and exemplified further below, the problems solved by the technique disclosed herein are relevant for any modern electronic device equipped with a camera **15** and suitable to be implemented in any such device. This is so since any modern electronic device of the kind specified (i.e. a device including an integral camera unit) is to be as slim as possible, as light as possible, and is to acquire pictures with as good quality as possible.

Modern cameras typically require zooming functions. When such a camera is used in an electronic device, such as a mobile phone device, the zooming function is often implemented with static optics. The problems which may arise when trying to incorporate Wide and Tele lenses into a common housing due to the difference in their heights are described above with reference to FIG. 1A.

As mentioned above, the presently disclosed subject matter includes a novel mobile electronic device **10** which includes an integrated camera unit **15** which is mounted inside the device casing **14**. The camera **15** includes at least one telephoto lens unit (not shown here) which is made of polymer materials. The telephoto lens unit is configured such that its total track lens (TTL) is less than 15 mm and even less than 10 mm, e.g. less than 6 mm or even less than 4 mm. Thus, enabling the camera to be fully integrated in the portable device (substantially not protruding from the device casing).

Reference is made to FIG. 1C showing schematically the configuration of a telephoto lens unit **20** of the present invention. The telephoto lens unit **20** is composed of multiple lens elements made of different polymer materials, i.e. materials having different Abbe numbers. The multiple lens elements are configured and arranged to define a telephoto lens assembly **22A** and a field lens assembly **22B** arranged along an optical axis OA with a predetermined effective gap G between them (as will be described more specifically further below). The telephoto lens assembly **22A** is configured to provide the telephoto optical effect of the telephoto lens unit **20**. The field lens assembly **22B** spaced from the telephoto lens assembly **22A** by the predetermined effective gap G is configured for correcting field curvature of the telephoto lens assembly **22A** and to compensate for residual chromatic aberrations of the telephoto lens assembly dispersed during light passage through the effective gap G.

The telephoto lens unit **20** is characterized by a total track lens (TTL) and an effective focal lens (EFL) such that $TTL < EFL$. This will be exemplified further below. According to the invention, the effective gap G between assemblies **22A** and **22B** is selected to be larger than $TTL/5$ of the telephoto lens unit **22A**, thereby enabling correction of field curvature of telephoto lens assembly **22A** by the field lens assembly **22B**.

The telephoto lens assembly **22A** includes three lens elements (generally three or more) L1, L2, L3 (which are

shown here schematically and not to scale), where lens **L1** has positive optical power and lenses **L2** and **L3** have together negative optical power. Lenses **L2** and **L3** are made of the first polymer material having a first Abbe number selected for reducing chromatic aberrations of the telephoto lens assembly **22A**. The field lens assembly **22B** includes two (or more) lens elements **L4** and **L5** which are made of different polymer materials respectively having different Abbe numbers. These lenses are configured to compensate for residual chromatic aberrations of the telephoto lens assembly **22A** dispersed during light passage through the effective gap **G** between the **22A** and **22B**.

Lenses **L1-L5** can be made for example of two plastic materials, one having an Abbe number greater than 50 and the other-smaller than 30. For example, Lenses **L1**, **L3** and **L5** are made of plastic with an Abbe number greater than 50, and lenses **L2** and **L4** are made of plastic having an Abbe number smaller than 30.

The following are several specific, but non-limiting, examples of the implementation and operation of the telephoto lens unit of the invention described above with reference to FIG. 1C. In the following description, the shape (convex or concave) of a lens element surface is defined as viewed from the respective side (i.e. from an object side or from an image side).

FIG. 2A shows a schematic illustration of an optical lens unit **100**, according to a first example of the presently disclosed subject matter. FIG. 2B shows the MTF vs. focus shift of the entire optical lens unit for various fields in the lens unit configuration **100**. FIG. 2C shows the distortion +Y in percent vs. field.

According to the example illustrated in FIG. 2A, lens unit **100** includes, in order from an object side to an image side, a first plastic lens element **102** (also referred to as “**L1**”) with positive refractive power having a convex object-side surface **102a** and a convex or concave image-side surface **102b**; a second plastic lens element **104** (also referred to as “**L2**”) with negative refractive power and having a meniscus convex object-side surface **104a**, with an image side surface marked **104b**; a third plastic lens element **106** (also referred to as “**L3**”) with negative refractive power having a concave object-side surface **106a** with an inflection point and a concave image-side surface **106b**. These lens elements define together the telephoto lens assembly (**22A** in FIG. 1C). Further provided in lens unit **100** is a fourth plastic lens element **108** (also referred to as “**L4**”) with positive refractive power having a positive meniscus, with a concave object-side surface marked **108a** and an image-side surface marked **108b**; and a fifth plastic lens element **110** (also referred to as “**L5**”) with negative refractive power having a negative meniscus, with a concave object-side surface marked **110a** and an image-side surface marked **110b**. These two lenses define together the field lens assembly (**22B** in FIG. 1C). The optical lens unit **100** may further optionally include a stop element **101**. The telephoto lens unit **100** defines an image plane **114** in which image sensor(s) is/are located, which is not shown here. Also, as exemplified in the figure, an optional glass window **112** is disposed between the image-side surface **110b** of fifth lens element **110** and the image plane **114**.

In the example of the telephoto lens unit **100**, all lens element surfaces are aspheric. Detailed optical data is shown in Table 1, and aspheric surface data is shown in Table 2, wherein the units of the radius of curvature (R), lens element thickness and/or distances between elements along the optical axis and diameter are expressed in mm. “Nd” is the refraction index. The equation of the aspheric surface profiles is expressed by:

$$z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \alpha_1 r^2 + \alpha_2 r^4 + \alpha_3 r^6 + \alpha_4 r^8 + \alpha_5 r^{10} + \alpha_6 r^{12} + \alpha_7 r^{14}$$

where r is the distance from (and is perpendicular to) the optical axis, k is the conic coefficient, $c=1/R$ where R is the radius of curvature, and α are coefficients given in Table 2.

In the equation above as applied to the telephoto lens unit, coefficients α_1 and α_7 are zero. It should be noted that the maximum value of r “max r”=Diameter/2. It should also be noted that in Table 1 (and in Tables 3 and 5 below), the distances between various elements (and/or surfaces) are marked “Lmn” (where m refers to the lens element number, n=1 refers to the element thickness and n=2 refers to the air gap to the next element) and are measured on the optical axis z, wherein the stop is at z=0. Each number is measured from the previous surface. Thus, the first distance -0.466 mm is measured from the stop to surface **102a**, the distance L11 from surface **102a** to surface **102b** (i.e. the thickness of first lens element **102**) is 0.894 mm, the air gap L12 between surfaces **102b** and **104a** is 0.020 mm, the distance L21 between surfaces **104a** and **104b** (i.e. thickness d2 of second lens element **104**) is 0.246 mm, etc. Also, L21=d₂ and L51=d₅. The lens elements in Tables 1 and 2 (as well as in Tables 3-6) are designed to provide an image on an entire 1/3” sensor having dimensions of approximately 4.7×3.52 mm. The optical diameter in all of these lens assemblies is the diameter of the second surface of the fifth lens element.

TABLE 1

#	Comment	Radius R [mm]	Distances [mm]	Nd/Vd	Diameter [mm]
1	Stop	Infinite	-0.466		2.4
2	L11	1.5800	0.894	1.5345/57.095	2.5
3	L12	-11.2003	0.020		2.4
4	L21	33.8670	0.246	1.63549/23.91	2.2
5	L22	3.2281	0.449		1.9
6	L31	-12.2843	0.290	1.5345/57.095	1.9
7	L32	7.7138	2.020		1.8
8	L41	-2.3755	0.597	1.63549/23.91	3.3
9	L42	-1.8801	0.068		3.6
10	L51	-1.8100	0.293	1.5345/57.095	3.9
11	L52	-5.2768	0.617		4.3
12	Window	Infinite	0.210	1.5168/64.17	3.0
13		Infinite	0.200		3.0

TABLE 2

#	Conic coefficient k	α_2	α_3	α_4	α_5	α_6
2	-0.4668	7.9218E-03	2.3146E-02	-3.3436E-02	2.3650E-02	-9.2437E-03
3	-9.8525	2.0102E-02	2.0647E-04	7.4394E-03	-1.7529E-02	4.5206E-03

TABLE 2-continued

#	Conic coefficient k	α_2	α_3	α_4	α_5	α_6
4	10.7569	-1.9248E-03	8.6003E-02	1.1676E-02	-4.0607E-02	1.3545E-02
5	1.4395	5.1029E-03	2.4578E-01	-1.7734E-01	2.9848E-01	-1.3320E-01
6	0.0000	2.1629E-01	4.0134E-02	1.3615E-02	2.5914E-03	-1.2292E-02
7	-9.8953	2.3297E-01	8.2917E-02	-1.2725E-01	1.5691E-01	-5.9624E-02
8	0.9938	-1.3522E-02	-7.0395E-03	1.4569E-02	-1.5336E-02	4.3707E-03
9	-6.8097	-1.0654E-01	1.2933E-02	2.9548E-04	-1.8317E-03	5.0111E-04
10	-7.3161	-1.8636E-01	8.3105E-02	-1.8632E-02	2.4012E-03	-1.2816E-04
11	0.0000	-1.1927E-01	7.0245E-02	-2.0735E-02	2.6418E-03	-1.1576E-04

Lens unit **100** provides a field of view (FOV) of 44 degrees, with EFL=6.90 mm, F #=2.80 and TTL of 5.904 mm. Thus and advantageously, the ratio TTL/EFL=0.855. Advantageously, the Abbe number of the first, third and fifth lens element is 57.095. Advantageously, the first air gap between lens elements **102** and **104** (the gap between surfaces **102b** and **104a**) has a thickness (0.020 mm) which is less than a tenth of thickness d_2 (0.246 mm). Advantageously, the Abbe number of the second and fourth lens elements is 23.91. Advantageously, an effective third air gap G (see below with reference to Table 9) between lens elements **106** and **108** (i.e. the telephoto and field lens assemblies) is greater than TTL/5. Advantageously, an effective fourth air gap (see below with reference to Table 9) between lens elements **108** and **110** is smaller than TTL/50.

The focal length (in mm) of each lens element in lens unit **100** is as follows: $f_1=2.645$, $f_2=-5.578$, $f_3=-8.784$, $f_4=9.550$ and $f_5=-5.290$. The condition $1.2 \times |f_3| > |f_2| > 1.5 \times f_1$ is clearly satisfied, as $1.2 \times 8.787 > 5.578 > 1.5 \times 2.645$. f_1 also fulfills the condition $f_1 < \text{TTL}/2$, as $2.645 < 2.952$.

FIG. 3A shows a schematic illustration of an optical lens unit **200**, according to another example of the presently disclosed subject matter. FIG. 3B shows the MTF vs. focus shift of the entire optical lens system for various fields in embodiment **200**. FIG. 3C shows the distortion +Y in percent vs. field.

According to the example illustrated in FIG. 3A, lens unit **200** comprises, in order from an object side to an image side: an optional stop **201**; a telephoto lens assembly including a first plastic lens element **202** with positive refractive power having a convex object-side surface **202a** and a convex or concave image-side surface **202b**, a second plastic lens

power having a positive meniscus, with a concave object-side surface marked **208a** and an image-side surface marked **208b**, and a fifth plastic lens element **210** with negative refractive power having a negative meniscus, with a concave object-side surface marked **110a** and an image-side surface marked **210b**. The optical lens unit **200** further optionally includes a glass window **212** disposed between the image-side surface **210b** of fifth lens element **210** and an image plane **214**.

In the lens unit **200**, all lens element surfaces are aspheric. Detailed optical data is given in Table 3, and the aspheric surface data is given in Table 4, wherein the markings and units are the same as in, respectively, Tables 1 and 2. The equation of the aspheric surface profiles is the same as for lens unit **100** described above.

TABLE 3

#	Comment	Radius R [mm]	Distances [mm]	Nd/Vd	Diameter [mm]
1	Stop	Infinite	-0.592		2.5
2	L11	1.5457	0.898	1.53463/56.18	2.6
3	L12	-127.7249	0.129		2.6
4	L21	6.6065	0.251	1.91266/20.65	2.1
5	L22	2.8090	0.443		1.8
6	L31	9.6183	0.293	1.53463/56.18	1.8
7	L32	3.4694	1.766		1.7
8	L41	-2.6432	0.696	1.632445/23.35	3.2
9	L42	-1.8663	0.106		3.6
10	L51	-1.4933	0.330	1.53463/56.18	3.9
11	L52	-4.1588	0.649		4.3
12	Window	Infinite	0.210	1.5168/64.17	5.4
13		Infinite	0.130		5.5

TABLE 4

#	Conic coefficient k	α_2	α_3	α_4	α_5	α_6
2	0.0000	-2.7367E-03	2.8779E-04	-4.3661E-03	3.0069E-03	-1.2282E-03
3	-10.0119	4.0790E-02	-1.8379E-02	2.2562E-02	-1.7706E-02	4.9640E-03
4	10.0220	4.6151E-02	5.8320E-02	-2.0919E-02	-1.2846E-02	8.8283E-03
5	7.2902	3.6028E-02	1.1436E-01	-1.9022E-02	4.7992E-03	-3.4079E-03
6	0.0000	1.6639E-01	5.6754E-02	-1.2238E-02	-1.8648E-02	1.9292E-02
7	8.1261	1.5353E-01	8.1427E-02	-1.5773E-01	1.5303E-01	-4.6064E-02
8	0.0000	-3.2628E-02	1.9535E-02	-1.6716E-02	-2.0132E-03	2.0112E-03
9	0.0000	1.5173E-02	-1.2252E-02	3.3611E-03	-2.5303E-03	8.4038E-04
10	-4.7688	-1.4736E-01	7.6335E-02	-2.5539E-02	5.5897E-03	-5.0290E-04
11	0.00E+00	-8.3741E-02	4.2660E-02	-8.4866E-03	1.2183E-04	7.2785E-05

element **204** with negative refractive power, having a meniscus convex object-side surface **204a**, with an image side surface marked **204b**, and a third plastic lens element **206** with negative refractive power having a concave object-side surface **206a** with an inflection point and a concave image-side surface **206b**; and a field lens assembly including a fourth plastic lens element **208** with positive refractive

Lens unit **200** provides a FOV of 43.48 degrees, with EFL=7 mm, F #=2.86 and TTL=5.90 mm. Thus, advantageously, the ratio TTL/EFL=0.843. Advantageously, the Abbe number of the first, third and fifth lens elements is 56.18. The first air gap between lens elements **202** and **204** has a thickness (0.129 mm) which is about half the thickness d_2 (0.251 mm). Advantageously, the Abbe number of the

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second lens element is 20.65 and of the fourth lens element is 23.35. Advantageously, the effective third air gap G between lens elements **206** and **208** is greater than TTL/5. Advantageously, the effective fourth air gap between lens elements **208** and **210** is smaller than TTL/50.

The focal length (in mm) of each lens element in lens unit **200** is as follows: f1=2.851, f2=-5.468, f3=-10.279, f4=7.368 and f5=-4.536. The condition $1.2 \times |f3| > |f2| > 1.5 \times f1$ is clearly satisfied, as $1.2 \times 10.279 > 5.468 > 1.5 \times 2.851$. f1 also fulfills the condition $f1 < TTL/2$, as $2.851 < 2.950$.

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TABLE 5-continued

#	Comment	Radius R [mm]	Distances [mm]	Nd/Vd	Diameter [mm]
7	L32	4.5841	1.998		1.7
8	L41	-2.8827	0.514	1.63549/23.91	3.4
9	L42	-1.9771	0.121		3.7
10	L51	-1.8665	0.431	1.5345/57.09	4.0
11	L52	-6.3670	0.538		4.4
12	Window	Infinite	0.210	1.5168/64.17	3.0
13		Infinite	0.200		3.0

TABLE 6

#	Conic coefficient k	α_2	α_3	α_4	α_5	α_6
2	-0.534	1.3253E-02	2.3699E-02	-2.8501E-02	1.7853E-02	-4.0314E-03
3	-13.473	3.0077E-02	4.7972E-03	1.4475E-02	-1.8490E-02	4.3565E-03
4	-10.132	7.0372E-04	1.1328E-01	1.2346E-03	-4.2655E-02	8.8625E-03
5	5.180	-1.9210E-03	2.3799E-01	-8.8055E-02	2.1447E-01	-1.2702E-01
6	0.000	2.6780E-01	1.8129E-02	-1.7323E-02	3.7372E-02	-2.1356E-02
7	10.037	2.7660E-01	-1.0291E-02	-6.0955E-02	7.5235E-02	-1.6521E-02
8	1.703	2.6462E-02	-1.2633E-02	-4.7724E-04	-3.2762E-03	1.6551E-03
9	-1.456	5.7704E-03	-1.8826E-02	5.1593E-03	-2.9999E-03	8.0685E-04
10	-6.511	-2.1699E-01	1.3692E-01	-4.2629E-02	6.8371E-03	-4.1415E-04
11	0.000	-1.5120E-01	8.6614E-02	-2.3324E-02	2.7361E-03	-1.1236E-04

FIG. 4A shows a schematic illustration of an optical lens unit **300**, according to yet a further example of the presently disclosed subject matter. FIG. 4B shows the MTF vs. focus shift of the entire optical lens system for various fields in embodiment **300**. FIG. 4C shows the distortion +Y in percent vs. field.

Lens unit **300** comprises, in order from an object side to an image side, an optional stop **301**; a telephoto lens assembly including a first plastic lens element **302** with positive refractive power having a convex object-side surface **302a** and a convex or concave image-side surface **302b**, a second plastic lens element **204** with negative refractive power, having a meniscus convex object-side surface **304a**, with an image side surface marked **304b**, a third plastic lens element **306** with negative refractive power having a concave object-side surface **306a** with an inflection point and a concave image-side surface **306b**; and a field lens assembly including a fourth plastic lens element **308** with positive refractive power having a positive meniscus, with a concave object-side surface marked **308a** and an image-side surface marked **308b**, and a fifth plastic lens element **310** with negative refractive power having a negative meniscus, with a concave object-side surface marked **310a** and an image-side surface marked **310b**. Also, an optional glass window **312** may be disposed between the image-side surface **310b** of fifth lens element **310** and an image plane **314**.

According to the present example of lens unit **300**, all lens element surfaces are aspheric. Detailed optical data is given in Table 5, and the aspheric surface data is given in Table 6, wherein the markings and units are the same as in, respectively, Tables 1 and 2. The equation of the aspheric surface profiles is the same as for lens units **100** and **200**.

TABLE 5

#	Comment	Radius R [mm]	Distances [mm]	Nd/Vd	Diameter [mm]
1	Stop	Infinite	-0.38		2.4
2	L11	1.5127	0.919	1.5148/63.1	2.5
3	L12	-13.3831	0.029		2.3
4	L21	8.4411	0.254	1.63549/23.91	2.1
5	L22	2.6181	0.426		1.8
6	L31	-17.9618	0.265	1.5345/57.09	1.8

Lens unit **300** provides a FOV of 44 degrees, EFL=6.84 mm, F #=2.80 and TTL=5.904 mm. Thus, advantageously, the ratio TTL/EFL=0.863. Advantageously, the Abbe number of the first lens element is 63.1, and of the third and fifth lens elements is 57.09. The first air gap between lens elements **302** and **304** has a thickness (0.029 mm) which is about $1/10^{th}$ the thickness d_2 (0.254 mm). Advantageously, the Abbe number of the second and fourth lens elements is 23.91. Advantageously, the effective third air gap G between lens elements **306** and **308** is greater than TTL/5. Advantageously, the effective fourth air gap between lens elements **308** and **310** is smaller than TTL/50.

The focal length (in mm) of each lens element in embodiment **300** is as follows: f1=2.687, f2=-6.016, f3=-6.777, f4=8.026 and f5=-5.090. The condition $1.2 \times |f3| > |f2| > 1.5 \times f1$ is clearly satisfied, as $1.2 \times 6.777 > 6.016 > 1.5 \times 2.687$. f1 also fulfills the condition $f1 < TTL/2$, as $2.687 < 2.952$.

Tables 7 and 8 provide respectively detailed optical data and aspheric surface data for a fourth embodiment of an optical lens system disclosed herein. The markings and units are the same as in, respectively, Tables 1 and 2. The equation of the aspheric surface profiles is the same as for lens systems **100**, **200** and **300**. The lens elements in Tables 7 and 8 are designed to provide an image on an entire $1/4''$ sensor having dimensions of approximately 3.66×2.75 mm.

TABLE 7

#	Comment	Radius R [mm]	Distances [mm]	Nd/Vd	Diameter [mm]
1	Stop	Infinite	-0.427		2.1
2	L11	1.3860	0.847	1.534809/55.66	2.2
3	L12	-8.5270	0.073		2.1
4	L21	11.1443	0.239	1.639078/23.253	1.9
5	L22	1.8641	0.504		1.7
6	L31	19.7342	0.239	1.534809/55.66	1.7
7	L32	3.9787	1.298		1.7
8	L41	-3.3312	0.522	1.639078/23.253	2.8
9	L42	-1.7156	0.079		3.1
10	L51	-1.7788	0.298	1.534809/55.66	3.5
11	L52	-12.6104	0.792		3.7

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TABLE 7-continued

#	Comment	Radius R [mm]	Distances [mm]	Nd/Vd	Diameter [mm]
12	Window	Infinite	0.210	1.5168/64.17	4.5
13		Infinite	0.177		4.6

TABLE 8

#	Conic coefficient k	$\alpha 2$	$\alpha 3$	$\alpha 4$	$\alpha 5$	$\alpha 6$
2	-0.326	8.776E-03	2.987E-02	-6.001E-02	6.700E-02	-2.849E-02
3	-10.358	4.266E-02	-2.240E-02	2.914E-02	-3.025E-02	3.108E-03
4	11.447	-3.257E-02	9.780E-02	-1.143E-02	-3.844E-02	1.005E-02
5	-0.026	-3.631E-02	2.928E-01	-2.338E-01	3.334E-01	-2.760E-02
6	0.000	1.578E-01	-2.229E-02	-4.991E-02	1.663E-01	-1.298E-01
7	3.860	2.044E-01	5.451E-02	-3.199E-01	5.619E-01	-3.663E-01
8	4.094	3.706E-02	-5.931E-02	4.662E-02	-4.654E-02	1.606E-02
9	-9.119	-7.980E-02	-1.376E-03	5.622E-03	-6.715E-03	2.127E-03
10	-12.777	-2.695E-01	1.894E-01	-5.690E-02	8.689E-03	-5.269E-04
11	0.000	-1.807E-01	1.278E-01	-4.504E-02	6.593E-03	-2.357E-04

The focal length (in mm) of each lens element according to this example is as follows: $f1=2.298$, $f2=-3.503$, $f3=-9.368$, $f4=4.846$ and $f5=-3.910$. The condition $1.2 \times |f3| > |f2| > 1.5 \times f1$ is clearly satisfied, as $1.2 \times 9.368 > 3.503 > 1.5 \times 2.298$. $f1$ also fulfills the condition $f1 < TTL/2$, as $2.298 < 2.64$.

Generally, with regard to the effective air gap between the adjacent lens elements, the following should be noted.

In each one of the lens units exemplified above, the first three lens elements (L1, L2 and L3) achieve essentially a telephoto effect for all fields (angles of object orientation relative to the optical axis), i.e. achieve a strong concentration (by L1) followed by partial collimation (mainly by L2 but also by L3). The fact that all fields need to have essentially the same telephoto effect leads to relatively small distances (small air gaps) between the three lens elements, e.g. especially between L1 and L2 (air gap 1). L4 and L5 are mainly field lens elements for reducing field curvature, i.e. their main effect is to cause the focal point for all fields (where the object distance is approximately infinity) to reside on the sensor plane. To achieve this, it is advantageous that for every field, the corresponding rays hit L4 and L5 at different locations, thus enabling separate adjustment for every field ("field separation").

The inventors have found that the desired fields' separation is obtainable in a lens unit design characterized by an "effective air gap" G between lenses L3 and L4 (between the telephoto and field lens assemblies, where a larger G leads to larger separation between the fields).

FIG. 5 illustrates the concept of the effective air gap between the two adjacent lens elements. First, an "air gap per field" D_{f-n} is defined as the length of the n^{th} field's chief ray along the respective chief ray between adjacent lens elements. Effective gap D_{Leff} is then defined as the average of N air gaps per field for field angles α separated evenly between $\alpha=0$ (for ray 1, air gap D_{f-1}) to $\alpha=\alpha_{max}$ (for ray N, air gap D_{f-N}), where ray N hits the end pixel on the image sensor diagonal. In other words, between each pair of adjacent lens elements (e.g. between L3 and L4 and between L4 and L5):

$$D_{Leff} = \left(\sum_{n=1}^N D_{f-n} \right) / N$$

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In essence, the effective air gap between adjacent lens elements reflects an average effective distance between the two surfaces bounding the air gap between the two adjacent lens elements. Exemplarily, in FIG. 5 there are N=9 chief rays (and 9 related field air gaps) and the chief rays are distributed angularly evenly between $\alpha=0$ for ray 1 and α_{max}

for ray 9. At α_{max} , ray 9 hits the end pixel on the image sensor diagonal.

Table 9 shows data on TTL, D_{Leff-3} , D_{Leff-4} and ratios between the TTL and the effective air gaps for each of lens units 100, 200 and 300 above D_{Leff-3} and D_{Leff-4} were calculated using 9 chief rays, as shown in FIG. 4.

TABLE 9

Embodi- ment	TTL	$D_{Leff-3} = G$	D_{Leff-4}	D_{Leff-3}/TTL	D_{Leff-4}/TTL
100	5.903	1.880	0.086	0.319	0.015
200	5.901	1.719	0.071	0.291	0.012
300	5.904	1.925	0.094	0.326	0.016
400	5.279	1.263	0.080	0.246	0.015

Using $D_{Leff-3}=G$ instead of the commonly used distance along the optical axis between L3 and L4 ensures better operation (for the purpose of reduction of field curvature) of lens elements L4 and L5 for all the fields. As seen in Table 9, good field separation may exemplarily be achieved if $D_{Leff-3}=G > TTL/5$.

A compact optical design requires that the diameter of L5 be as small as possible while providing the required performance. Since the lens and camera footprint is determined by L5 diameter, a small effective air gap, D_{Leff-4} , between lenses L4 and L5 is advantageous in that it allows a small diameter of lens L5 without degrading the optical performance. Effective air gap D_{Leff-4} is a better indicator of the L5 diameter than the commonly used air gap along the optical axis between L4 and L5. An adequately small L5 diameter may exemplarily be achieved if the effective air gap between the field lenses L4 and L5 is $D_{Leff-4} < TTL/50$. It should be noted that an effective air gap D_{Leff} can be calculated in principle using any combination of two or more chief rays (for example ray 1 and ray 9 in FIG. 4). However, the "quality" of D_{Leff} calculation improves while considering an increased number of chief rays.

The miniature telephoto lens units described above with reference to FIGS. 1C and 2 to 5 are designed with a TTL shorter than EFL. Accordingly, due to shorter TTL, such lens units have a smaller field of view, as compared to standard mobile phone lens units. Therefore, it would be particularly useful to use such a telephoto lens unit as a Tele sub-camera

lens unit in a dual aperture zoom camera. Such a dual aperture zoom camera is described in the above-mentioned WO14199338 of the same assignee as the present application.

As mentioned above, a problem associated with the use of conventional Wide and Tele lens modules in a camera is associated with the different lengths/heights of the lenses which can cause shadowing and light blocking effects. According to the presently disclosed subject matter it is suggested to eliminate or at least significantly reduce these shadowing and light blocking effects by replacing the conventional Tele lens module by the miniature telephoto lens unit described above in the dual aperture camera.

Thus, according to the presently disclosed subject matter, the problem discussed above posed by a difference in the TTL/EFL ratios of the conventional Tele and Wide lenses may be solved through use of a standard lens for the Wide camera ($TTL_W/EFL_W > 1.1$, typically 1.3) and of a special Telephoto lens design for the Tele camera ($TTL_T/EFL_T < 1$, e.g. 0.87), where the telephoto lens unit is configured as described above, providing the miniature telephoto lens unit.

Using the above described miniature telephoto lens unit enables to reduce the TTL_T (according to one non-limiting example down to $7 \times 0.87 = 6.09$ mm) leading to a camera height of less than 7 mm (which is an acceptable height for a smartphone or any other mobile electronic device). The height difference between the telephoto lens unit and the Wide lens unit is also reduced to approximately 1.65 mm, thus reducing shadowing and light blocking problems.

According to some examples of a dual-aperture camera disclosed herein, the ratio " e "= EFL_T/EFL_W is in the range 1.3-2.0. In some embodiments, the ratio $TTL_T/TTL_W < 0.8e$. In some embodiments, TTL_T/TTL_W is in the range 1.0-1.25. According to some examples disclosed herein, EFL_W may be in the range 2.5-6 mm and EFL_T may be in the range 5-12 mm.

Referring now to the figures, FIG. 6A shows schematically in perspective cross section an example of a dual-aperture zoom camera device 600. Camera device 600 includes two camera units 602 and 604. It should be understood that the two camera units may be associated with common or separate detectors (pixel matrix and their associated read out circuits). Thus, the two camera units are actually different in their optics, i.e. in the imaging channels defined by the wide and telephoto lens units. Each camera unit may be mounted on a separate PCB (respectively 605a and 605b) including the read out circuit, and includes a lens unit (respectively 606 and 608), and an image sensor including a pixel matrix (respectively 614 and 616), and an actuator (respectively 610 and 612) associated with a focusing mechanism. In this embodiment, the two PCBs lie in the same plane. It should be understood that in the embodiment where the readout circuits of the two imaging channels are in the same plane, a common PCB can be used, as will be described further below. The two camera units are connected by a case 618. For example, camera 602 includes a Wide lens unit and camera 604 includes a Telephoto lens unit, the TTL_T of the lens unit defining the respective camera height H. For example, the Wide and Telephoto lens units provide respectively main and auxiliary optical/imaging paths, enabling to use the main image for interpreting the auxiliary image data.

FIG. 6B shows schematically, in perspective cross, another example of a dual-aperture zoom camera 600' utilizing the principles of the invention. Camera 600' is generally similar to the above-described camera 600, and the common components are shown in the figure in a self-

explanatory manner and thus are not indicated by reference numbers. As in camera 600, in the camera 600', the camera unit 602 (e.g. a Wide lens camera) and camera unit 604 (e.g. a Telephoto lens camera) are mounted on separate PCBs (respectively 605a and 605b). However, in contrast with camera 600, in camera 600' the two PCBs lie in different planes. This enables the object side principal planes of the Wide and Telephoto lens units to be close one to the other, thus reducing the dependency of magnification factor in the two units on the object distance.

For example, camera dimensions for the cameras shown in FIGS. 6A and 6B may be as follows: a length L of the camera (in the Y direction) may vary between 13-25 mm, a width W of the camera (in the X direction) may vary between 6-12 mm, and a height H of the camera (in the Z direction, perpendicular to the X-Y plane) may vary between 4-12 mm. More specifically, considering a smartphone camera example disclosed herein, $L=18$ mm, $W=8.5$ mm and $H=7$ mm.

FIG. 7 shows schematically, in perspective cross section, yet another example of a dual-aperture zoom camera 700. Camera 700 is similar to cameras 600 and 600' in that it includes two camera units 702 and 704 with respective lens units 706 and 708, respective actuators 710 and 712 and respective image sensors 714 and 716. However, in camera 700, the two camera units 702 and 704 are mounted on a single (common) PCB 705. The mounting on a single PCB and the minimizing of a distance "d" between the two camera units minimizes and may even completely avoid camera movement (e.g. associated with mishaps such as drop impact). In general, the dimensions of camera 700 may be in the same range as those of cameras 600 and 600'. However, for the same sensors and optics, the footprint $W \times L$ and the weight of camera 700 are smaller than that of cameras 600 and 600'. Mishaps such as drop impact may cause a relative movement between the two cameras after system calibration, changing the pixel matching between the Tele and Wide images and thus preventing fast reliable fusion of the Tele and Wide images. Therefore, such effects should preferably be eliminated.

As described above, the high-quality imaging is also associated with the implementation of standard optical image stabilization (OIS) in such a dual-aperture zoom camera. Standard OIS compensates for camera tilt ("CT"), i.e., image blur, by a parallel-to-the image sensor (exemplarily in the X-Y plane) lens movement ("LMV"). The amount of LMV (in millimeters) needed to counter a given camera tilt depends on the camera lens EFL, according to the relation:

$$LMV = CT * EFL,$$

where "CT" is in radians and EFL is in mm.

Since the Wide and telephoto lens units have significantly different EFLs, both lenses cannot move together and achieve optimal tilt compensation for both of the respective camera units. More specifically, since the tilt is the same for both camera units, a movement that will compensate for the tilt for the Wide camera unit will be insufficient to compensate for the tilt for the Telephoto camera unit, and vice versa. Using separate OIS actuators for the two camera units respectively can achieve simultaneous tilt compensation for both of them, but the entire system would be complex and costly, which is undesirable for portable electronic devices.

In this connection, reference is made to FIG. 8 which shows an example of a dual-aperture zoom camera 800 (similar to the above-described camera 700) that includes two camera units 802 and 804 mounted either on a single

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PCB **805** (as shown in this example) or on separate PCBs. Each camera unit includes a lens unit (respectively **806** and **808**), an actuator (respectively **810** and **812**) and an image sensor (respectively **814** and **816**). The two actuators are rigidly mounted on a rigid base **818** that is flexibly connected to the PCB (or PCBs) through flexible elements **820**. Base **818** is movable by an OIS mechanism (not shown) controlled by an OIS controller **902** (shown in FIG. 9). The OIS controller **902** is coupled to, and receives camera tilt information from a tilt sensor (e.g. a gyroscope) **904** (FIG. 9). More details of an example of an OIS procedure as disclosed herein are given below with reference to FIG. 9. The two camera units are separated by a small distance “d”, e.g. 1 mm. This small distance between camera units also reduces the perspective effect enabling smoother zoom transition between the camera units.

As indicated above, the two image sensors **814** and **816** may be mounted on separate PCBs that are rigidly connected, thereby enabling adaptation of an OIS mechanism to other system configurations, for example those described above with reference to FIGS. 6A and 6B.

In some embodiments, and optionally, a magnetic shield plate may be used, e.g. as described in co-owned U.S. patent application Ser. No. 14/365,718 titled “Magnetic shielding between voice coil motors in a dual-aperture camera”, which is incorporated herein by reference in its entirety. Such a magnetic shield plate may be inserted in the gap (with width d) between the Wide and Tele camera units.

In general, the dimensions of camera **800** may be in the same range as those of cameras **600**, **600'** and **700**.

Reference is made to FIG. 9, which exemplifies the camera operation, utilizing a tilt sensor **904** which dynamically measures the camera tilt (which is the same for both the Wide and Tele camera units). As shown, an OIS controller **902** (electronic circuit including hardware/software components) is provided, which is coupled to the actuators of both camera units (e.g. through base **818**), and receives a CT input from the tilt sensor **904** and a user-defined zoom factor, and controls the lens movement of the two camera units to compensate for the tilt. The LMV is for example in the X-Y plane. The OIS controller **902** is configured to provide a LMV equal to $CT \cdot EFL_{ZF}$, where “ EFL_{ZF} ” is chosen according to the user-defined zoom factor, ZF. According to one example of an OIS procedure, when $ZF=1$, LMV is determined by the Wide camera unit's EFL_W (i.e. $EFL_{ZF}=EFL_W$ and $LMV=CT \cdot EFL_W$). Further, when $ZF>e$ (i.e. $ZF>EFL_T/EFL_W$), LMV is determined by the telephoto camera unit's EFL_T (i.e. $EFL_{ZF}=EFL_T$ and $LMV=CT \cdot EFL_T$). Further yet, for a ZF between 1 and e, the EFL_{ZF} may shift gradually from EFL_W to EFL_T according to $EFL_{ZF}=ZF \cdot EFL_W$.

Thus, the present invention provides a novel approach for configuring a camera device suitable for use in portable electronic devices, in particular smart phones. The present invention solves various problems associated with the requirements for physical parameters of such devices (weight, size), high image quality and zooming effects.

What is claimed is:

1. A dual-aperture camera, comprising:

a Wide camera having a Wide lens with a field of view FOV_W and having a total track length TTL_W and an effective focal length EFL_W , and a Telephoto camera

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comprising a Telephoto lens with a Telephoto field of view FOV_T , a total track length TTL_T and an effective focal length EFL_T ,

wherein the TTL_T includes a glass window,

wherein a ratio TTL_W/EFL_W is greater than 1,

wherein a ratio TTL_T/EFL_T is smaller than 1,

wherein the Telephoto lens has an optical axis and a F # smaller than 3.2 and includes, in order from an object plane to an image plane along the optical axis, a first lens element having positive optical power and two last lens elements, wherein one of the two last lens elements has an Abbe number greater than 50, wherein the other of the two last lens elements has an Abbe number smaller than 30, and wherein a height H of the dual-aperture camera measured along the optical axis is in the range of 4 mm-12 mm.

2. The dual-aperture camera of claim 1, wherein EFL_T is in the range $EFL_T=5-12$ mm.

3. The dual-aperture camera of claim 1, wherein F # is smaller than 2.9.

4. The dual-aperture camera of claim 1, wherein TTL_T/EFL_T is smaller than 0.9.

5. The dual-aperture camera of claim 1, wherein the Telephoto lens unit includes a pair of lens elements formed by a second lens element and a third lens element, and wherein the pair of lens elements has a combined negative optical power.

6. The dual-aperture camera of claim 1, wherein the first lens element of the Telephoto lens unit has a focal length f1 smaller than $TTL_T/2$.

7. The dual-aperture camera of claim 1, wherein the two last lens elements are spaced apart from one another by an effective air gap smaller than $TTL_T/50$.

8. The dual-aperture camera of claim 1, wherein the first, second and third lens elements have respective focal lengths f1, f2 and f3, and wherein the respective focal lengths satisfy the condition $1.2|f3|>|f2|>1.5f1$.

9. The dual-aperture camera of claim 1, wherein the Telephoto camera unit includes an image sensor with an image sensor size $1/4"$ to $1/3"$.

10. The dual-aperture camera of claim 1, wherein the Wide and Telephoto camera units are spaced from one another a distance of about 1 mm.

11. The dual-aperture camera of claim 1, wherein TTL_T is smaller than 5.5 mm.

12. The dual-aperture of claim 1, included in a mobile electronic device.

13. The dual-aperture camera of claim 12, wherein the mobile device is a smartphone.

14. The dual-aperture of claim 2, included in a mobile electronic device.

15. The dual-aperture camera of claim 14, wherein the mobile device is a smartphone.

16. The dual-aperture of claim 3, included in a mobile electronic device.

17. The dual-aperture camera of claim 16, wherein the mobile device is a smartphone.

18. The dual-aperture camera of claim 4, included in a mobile electronic device.

19. The dual-aperture camera of claim 18, wherein the mobile device is a smartphone.

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